

PH-214 Modern Physics Notes

Useful Math

Taylor series expansions:

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\tan x = \sum_{n=1}^{\infty} \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!} x^{2n-1} = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \dots$$

$$(1+x)^m = \sum_{n=0}^{\infty} \binom{m}{n} x^n = 1 + mx + \frac{m(m-1)}{2} x^2 + \frac{m(m-1)(m-2)}{6} x^3 + \dots$$

Note: For small x , higher order terms reduce to zero

Euler's Identity:

Use complex exponentials to manipulate complicated trig functions and complex amplitudes.

$$e^{ix} = \cos x + i \sin x$$

Visualizing propagating plane waves:

A 1D plane wave propagating in the $+z$ direction can be mathematically represented as:

$$\psi = f(kz - \omega t)$$

Where:

- $(kz - \omega t)$ [rad]:

The argument represents the **phase** of the wave. A change in phase simply moves the wave forward or backward in space and time, without changing its shape, there are a few ways to see this which we will look at later.

- k [rad/m]:

is the **wave number**, representing the spatial frequency of the wave, or how *spread out* the wave is in space. Since k scales the spatial variable z , a larger k means the wave advances through the same phase over a smaller change in z .

- ω [rad/s]:

is the **angular frequency**, representing the temporal frequency of the wave, or how *spread out* the wave is in time. As in rotational kinematics, ω scales time t , and a larger ω means the wave advances through the same phase over a smaller change in t .

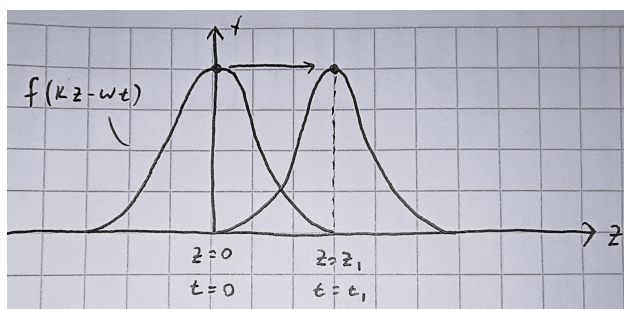
A plane wave is special in that it represents a wave whose shape is fixed as it propagates through space and time, with its shape represented by the function f .

Note:

$$k = \frac{2\pi}{\lambda}, \quad \frac{\omega}{2\pi} = f = \frac{1}{T} [\text{Hz}], \quad v = \frac{\omega}{k}$$

Global View

As mentioned before, there are two ways to visualize how the phase ($kz - \omega t$) represents a propagating wave. The first is a *global* view, looking at how z and t change relative to each other for a fixed phase, tracking a specific point on the wave.



When we track a specific point on the wave, f must give the same value at different positions and times, meaning the phase must stay constant. This shows that as t increases, z must also increase to maintain the phase (the wave moves rightward!). Vice versa, as z increases, t must increase too to keep the phase constant (again, moving rightward!).

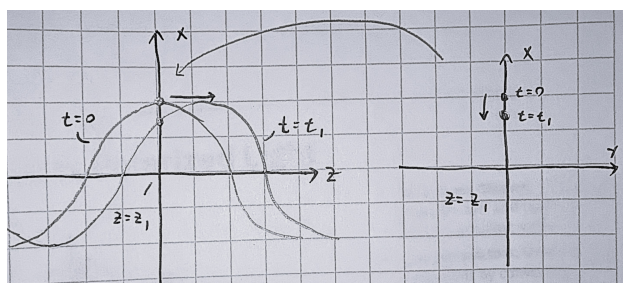
Note: This intuition applies to *any* point on the wave! Thus, by tracking all points, we can clearly see a propagating plane wave.

Local View

The second is a *local* view, looking at how the phase changes at a fixed position as time changes, or at a fixed time as position changes. For this, we will look at a sinusoidal wave, to better show oscillation: $\psi = \cos(kz - \omega t)$

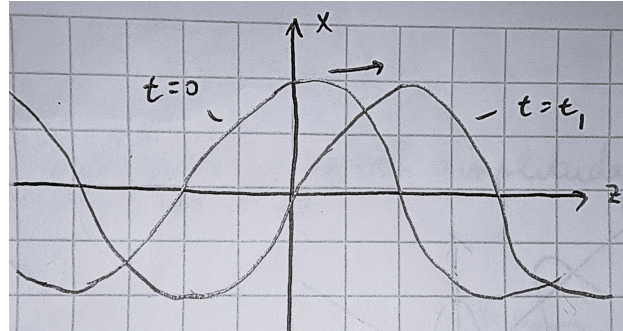
1. Fixing z , letting t vary:

Imagine we are at a fixed position $z = z_1$. As time increases, we are effectively watching a cross-section of the wave at that point, seeing only a single point along the wave moving up and down like riding the “surface” of the wave.



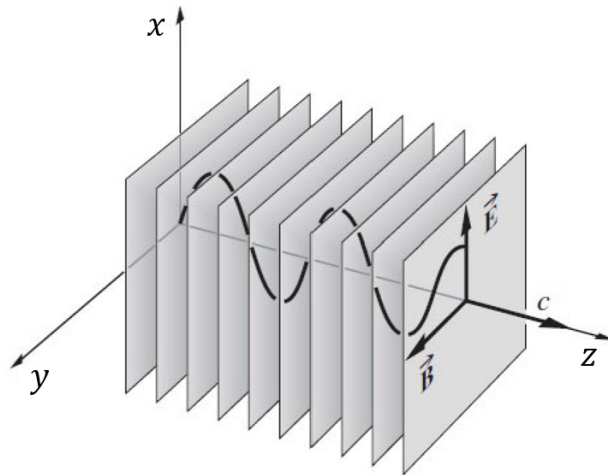
2. Fixing t , letting z vary:

Now imagine we are at a fixed time $t = t_1$. Moving along the z axis, and plotting the wave's amplitude, we see the full shape of the wave at that instant (ωt_1 becomes a constant phase shift).



3. Letting z and t vary together:

When we looked at z and t separately, we saw only a *part* of the wave's behavior. For a fixed z , we observed how a single point moves over time, capturing the wave's temporal variation. For a fixed t , we saw the spatial variation, revealing the full shape of the wave statically at that instant. When both z and t vary together, we can see the wave's complete dynamic behavior, watching how its shape propagates through space and time.



Note: This intuition will become very useful when we study electromagnetic polarization, where the plane wave is a 2D wave. In our cross-sectional view, the single point along the wave will move along the xy plane instead of just up and down.

Vector Calculus

- **Del Operator:**

$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$$

- **Laplacian Operator:**

$$\nabla^2 = \vec{\nabla} \cdot \vec{\nabla} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad (\text{in cartesian})$$

- **Divergence:** Dot product of del operator with a vector field $\vec{A}$, giving a scalar field that represents how much $\vec{A}$ spreads (diverges) out from a point.

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Note: A positive divergence indicates a source, while a negative divergence indicates a sink.

- **Curl:** Cross product of del operator with a vector field $\vec{A}$, giving a vector field that represents how much $\vec{A}$ curls around a point.

$$\begin{aligned} \vec{\nabla} \times \vec{A} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \\ &= \underbrace{\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right)}_{\text{curl about } x\text{-axis}} \hat{x} + \underbrace{\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right)}_{\text{curl about } y\text{-axis}} \hat{y} + \underbrace{\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)}_{\text{curl about } z\text{-axis}} \hat{z} \end{aligned}$$

Note: A positive curl indicates a counterclockwise rotation, while a negative curl indicates a clockwise rotation about the respective axis.

- **Vector Identities:**

Triple Products

$$\begin{aligned} \vec{A} \cdot (\vec{B} \times \vec{C}) &= \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B}) \\ \vec{A} \times (\vec{B} \times \vec{C}) &= \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B}) \end{aligned}$$

Product Rules

$$\begin{aligned} \vec{\nabla}(fg) &= f(\vec{\nabla}g) + g(\vec{\nabla}f) \\ \vec{\nabla}(\vec{A} \cdot \vec{B}) &= \vec{A} \times (\vec{\nabla} \times \vec{B}) + \vec{B} \times (\vec{\nabla} \times \vec{A}) + (\vec{A} \cdot \vec{\nabla})\vec{B} + (\vec{B} \cdot \vec{\nabla})\vec{A} \\ \vec{\nabla} \cdot (f\vec{A}) &= f(\vec{\nabla} \cdot \vec{A}) + \vec{A} \cdot (\vec{\nabla}f) \\ \vec{\nabla} \cdot (\vec{A} \times \vec{B}) &= \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B}) \\ \vec{\nabla} \times (f\vec{A}) &= f(\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla}f) \\ \vec{\nabla} \times (\vec{A} \times \vec{B}) &= (\vec{B} \cdot \vec{\nabla})\vec{A} - (\vec{A} \cdot \vec{\nabla})\vec{B} + \vec{A}(\vec{\nabla} \cdot \vec{B}) - \vec{B}(\vec{\nabla} \cdot \vec{A}) \end{aligned}$$

Second Derivatives

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0 \quad (\text{Divergence of a curl is always zero})$$

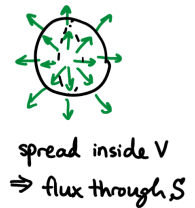
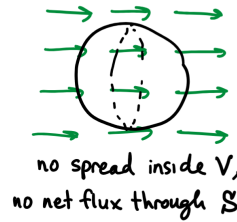
$$\vec{\nabla} \times (\vec{\nabla} f) = 0 \quad (\text{Curl of a gradient is always zero})$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} \quad (\text{Curl of a curl})$$

Divergence Theorem

Relates the flux of a vector field $\vec{F}$ through a closed surface S to the divergence of $\vec{F}$ within the volume V enclosed by S .

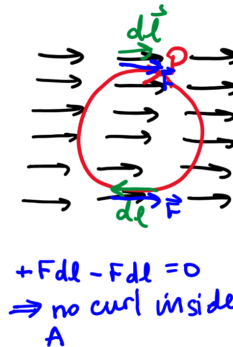
$$\underbrace{\int_V (\vec{\nabla} \cdot \vec{F}) d\tau}_{\text{how field spreads inside the volume}} = \underbrace{\oint_S \vec{F} \cdot d\vec{a}}_{\text{flux of field through the surface}}$$



Stokes Theorem

States that inside a surface A bounded by a closed curve P , if a vector field $\vec{F}$ has a non-zero curl about $\vec{A}$ (the normal to the surface), then there will be a non-zero circulation of $\vec{F}$ around the curve P .

$$\underbrace{\int_A (\vec{\nabla} \times \vec{F}) \cdot d\vec{a}}_{\text{flux of the curl of } \vec{F}} = \underbrace{\oint_P \vec{F} \cdot d\vec{l}}_{\text{circulation of } \vec{F}}$$



Key Electricity and Magnetism Laws

- Coulomb's Law

$$\vec{F}_e = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

- Magnetic Force

$$\vec{F}_b = q\vec{v} \times \vec{B}$$

- Electric Field of a Point Charge
sets up a diverging field

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

- Biot-Savart Law
sets up a curling field

$$\vec{B} = \int \frac{\mu_0 i d\vec{s} \times \hat{r}}{4\pi r^2}$$

Unit 1: Electromagnetic Waves

Maxwell's Equations

1. Gauss's Law for Electricity

Relates the electric flux through a closed 3D Gaussian surface to the total charge enclosed within that surface.

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc}}}{\epsilon_0}$$

There exists a diverging electric field if there exists a non-zero charge density at that point.

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

2. Gauss's Law for Magnetism

Any 3D Gaussian surface will have zero net magnetic flux (no magnetic monopoles).

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

There exists no diverging magnetic field at any point in space because there are no magnetic monopoles.

$$\vec{\nabla} \cdot \vec{B} = 0$$

3. Faraday's Law of Induction

Relates the electric circulation around a closed Faradian loop to the changing magnetic flux through the surface bounded by the loop.

$$\oint_P \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int_S \vec{B} \cdot d\vec{a}$$

Note: For coils with N turns, multiply the flux by N .

A changing magnetic field induces a circulating (curling) electric field.

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

4. Ampere-Maxwell Law

Relates the magnetic circulation around a closed Amperian loop to the enclosed current and changing electric flux through the surface bounded by the loop.

$$\oint_P \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}, \quad \Phi_E = \int_S \vec{E} \cdot d\vec{a}$$

Both a changing electric field and a non-zero current density induce a circulating (curling) magnetic field.

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}, \quad I_{\text{enc}} = \int_S \vec{J} \cdot d\vec{a}$$

Deriving the Differential Forms of Maxwell's Equations

By applying the divergence and stokes theorems to the integral forms of Maxwell's equations, we can derive their corresponding differential forms. This process allows us to express the behavior of electric and magnetic fields locally at every point in space and time, rather than just over closed surfaces or loops.

1. Gauss's Law for Electricity

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc}}}{\epsilon_0}$$

Applying the divergence theorem to the left-hand side:

$$\oint_S \vec{E} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{E}) d\tau$$

We can also write the enclosed charge as a volume integral of the charge density:

$$q_{\text{enc}} = \int_V \rho d\tau \quad \rho = (\text{vol charge density})$$

Thus,

$$\begin{aligned} \implies \int_V (\vec{\nabla} \cdot \vec{E}) d\tau &= \frac{1}{\epsilon_0} \int_V \rho d\tau \\ \implies \boxed{\vec{\nabla} \cdot \vec{E} &= \frac{\rho}{\epsilon_0}} \end{aligned}$$

2. Gauss's Law for Magnetism

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

Applying the divergence theorem:

$$\oint_S \vec{B} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{B}) d\tau = 0$$

Thus,

$$\implies \boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

3. Faraday's Law of Induction

$$\oint_P \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int_S \vec{B} \cdot d\vec{a}$$

Applying Stokes' theorem to the left-hand side:

$$\oint_P \vec{E} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a}$$

We can also write the changing magnetic flux as:

$$-\frac{d\Phi_B}{dt} = -\frac{d}{dt} \left[\int_S \vec{B} \cdot d\vec{a} \right] = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}$$

Note: Stokes theorem relates the circulation around a curve to the curl over a surface at a single instant in time. In other words, the surface S is *fixed* and does not change in time, so $d\vec{a}$ is *time independent* and the time derivative acts only on the integrand, allowing us to move the time derivative inside the integral.

Thus, equating the two sides gives:

$$\begin{aligned} \Rightarrow \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} &= - \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a} \\ \Rightarrow \boxed{\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}} \end{aligned}$$

4. Ampere-Maxwell Law

$$\oint_P \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}, \quad \Phi_E = \int_S \vec{E} \cdot d\vec{a}$$

Applying Stokes' theorem to the left-hand side:

$$\oint_P \vec{B} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a}$$

We can also write the enclosed current as a surface integral of the current density:

$$I_{\text{enc}} = \int_S \vec{J} \cdot d\vec{a}, \quad \vec{J} = (\text{area current density})$$

And the changing electric flux as:

$$\frac{d\Phi_E}{dt} = \frac{d}{dt} \left[\int_S \vec{E} \cdot d\vec{a} \right] = \int_S \frac{\partial \vec{E}}{\partial t} \cdot d\vec{a}$$

Thus, equating the two sides gives:

$$\begin{aligned} \Rightarrow \int_S (\vec{\nabla} \times \vec{B}) \cdot d\vec{a} &= \int_S \left(\mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{a} \\ \Rightarrow \boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}} \end{aligned}$$

Applying Vector Identities

Additionally, we can apply the vector identities we have to the differential forms of Maxwell's equations to derive some useful results.

Applying divergence of a curl is zero to Faraday's law:

$$\begin{aligned}\vec{\nabla} \cdot (\vec{\nabla} \times \vec{E}) &= 0 \\ \vec{\nabla} \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) &= 0 \\ -\frac{\partial}{\partial t} (\underbrace{\nabla \cdot \vec{B}}_{\text{Gauss's law for magnetism (=0)}}) &= 0 \\ \implies 0 &= 0\end{aligned}$$

Applying divergence of a curl is zero to Ampere-Maxwell law:

$$\begin{aligned}\vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) &= 0 \\ \vec{\nabla} \cdot \left(\mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) &= 0 \\ \mu_0 (\vec{\nabla} \cdot \vec{J}) + \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\underbrace{\nabla \cdot \vec{E}}_{\text{Gauss's law for electricity } (= \frac{\rho}{\epsilon_0})}) &= 0 \\ \mu_0 (\vec{\nabla} \cdot \vec{J}) + \mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\frac{\rho}{\epsilon_0} \right) &= 0 \\ \implies \boxed{\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0} &\quad (\text{Continuity Equation})\end{aligned}$$

We can rewrite the continuity equation into an integral form:

Integrating both sides over a volume V :

$$\int_V (\vec{\nabla} \cdot \vec{J}) d\tau = \int_V -\frac{\partial \rho}{\partial t} d\tau$$

Apply divergence theorem to the left-hand side:

$$\int_V (\vec{\nabla} \cdot \vec{J}) d\tau = \oint_S \vec{J} \cdot d\vec{a}$$

Since the volume V is fixed in space, we can pull the time derivative outside the integral.

We also know that the total charge enclosed is $q = \int_V \rho d\tau$:

$$\int_V -\frac{\partial \rho}{\partial t} d\tau = -\frac{d}{dt} \underbrace{\int_V \rho d\tau}_{q_{\text{enclosed}}} = -\frac{dq}{dt}$$

Thus, equating the two sides gives:

$$\oint_S \vec{J} \cdot d\vec{a} = -\frac{dq}{dt}$$

Note: The continuity equation is also called a flux based conservation law. It represents a local **conservation of charge** and states that if the total charge enclosed in a volume is changing ($\frac{dq}{dt} \neq 0$), then there must be a net current flowing into or out of the volume ($\oint_S \vec{J} \cdot d\vec{a} \neq 0$) that accounts for that change.



If a balloon were filled with a certain amount of water and that quantity of water was decreasing...



It must mean that there is a flux of water, passing through the surface, leaving the volume!

Showing that E and B Fields Exist as Waves in a Vacuum

In a vacuum, there are no charges or currents, so we can set $\rho = 0$ and $\vec{J} = \vec{0}$ in Maxwell's equations. This simplifies them to:

$\vec{\nabla} \cdot \vec{E} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$	no divergent fields (no source charge)
$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$	only exist curling fields induced by one another!

By applying the curl of a curl rule to Faraday's and Ampere-Maxwell laws, we can derive the wave equations for the electric and magnetic fields in a vacuum.

$\begin{aligned} \vec{\nabla} \times (\vec{\nabla} \times \vec{E}) &= \underbrace{\vec{\nabla}(\vec{\nabla} \cdot \vec{E})}_{0 \text{ in a vacuum}} - \nabla^2 \vec{E} \\ \vec{\nabla} \times \left(-\frac{\partial \vec{B}}{\partial t} \right) &= -\nabla^2 \vec{E} \\ -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B}) &= -\nabla^2 \vec{E} \\ -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) &= -\nabla^2 \vec{E} \end{aligned}$	$\begin{aligned} \vec{\nabla} \times (\vec{\nabla} \times \vec{B}) &= \underbrace{\vec{\nabla}(\vec{\nabla} \cdot \vec{B})}_0 - \nabla^2 \vec{B} \\ \mu_0 \epsilon_0 \frac{\partial}{\partial t} \underbrace{(\vec{\nabla} \times \vec{E})}_{-\frac{\partial \vec{B}}{\partial t}} &= -\nabla^2 \vec{B} \\ -\mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} &= -\nabla^2 \vec{B} \end{aligned}$
$\Rightarrow \boxed{\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}}$	$\Rightarrow \boxed{\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}}$

Recall: The classic wave equation is

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad \Rightarrow \quad v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$$

Modeling EM Radiation (Light) in a Vacuum

Properties of light:

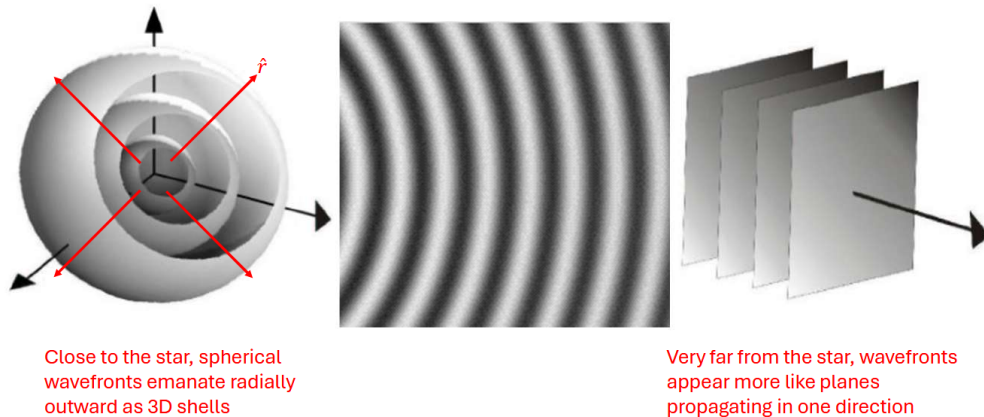
- $\vec{E} \times \vec{B}$ points in the direction of propagation.
- $\vec{E}$ and $\vec{B}$ are perpendicular to each other and to the direction of propagation (transverse wave).
- $\vec{E}$ and $\vec{B}$ oscillate in phase as to perpetually induce each other according to Faraday's and Ampere-Maxwell laws.
- The magnitudes of $\vec{E}$ and $\vec{B}$ are related by $E = cB$.
- Light does not require a medium to propagate.
- Light has speed

$$v = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Our goal is to find a solution for $\vec{E}$ and $\vec{B}$ that satisfies the 3D wave equations as well as Maxwell's equations in a vacuum.

Plane Wave Approximation

Consider the following wavefronts generated by a point source of light (ie. a star):

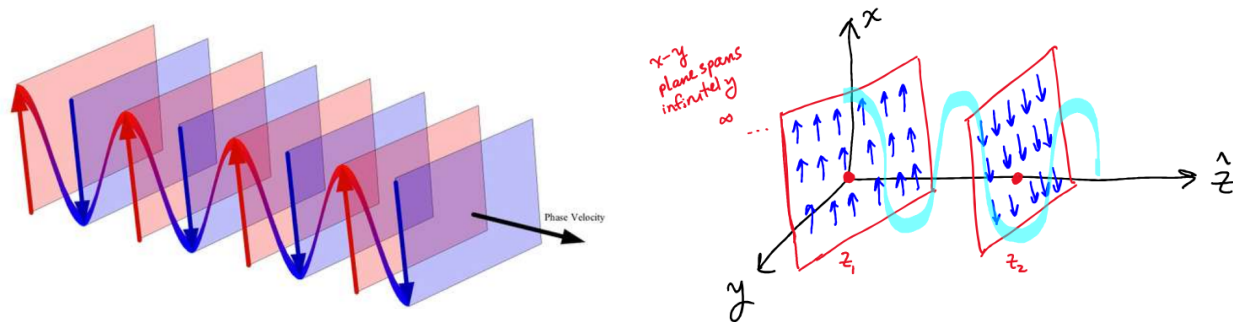


The radius of curvature of the wavefronts increases as they travel outward. Thus, at a far enough distance from the source, the wavefronts will appear to be flat planes consisting of electric and magnetic fields with *uniform* magnitude and direction. This is called the **plane wave approximation**.

The plane wave model also assumes **monochromatic light**, having a fixed wavelength λ and wavenumber k ($k = 2\pi/\lambda$). This also implies a fixed frequency ω , since the speed of light is constant:

$$v = \frac{\omega}{k} = c \quad \implies \quad \omega = ck.$$

Without a single frequency, the wavefronts would interfere with one another and would no longer form the ideal uniform planes described by the plane wave approximation.

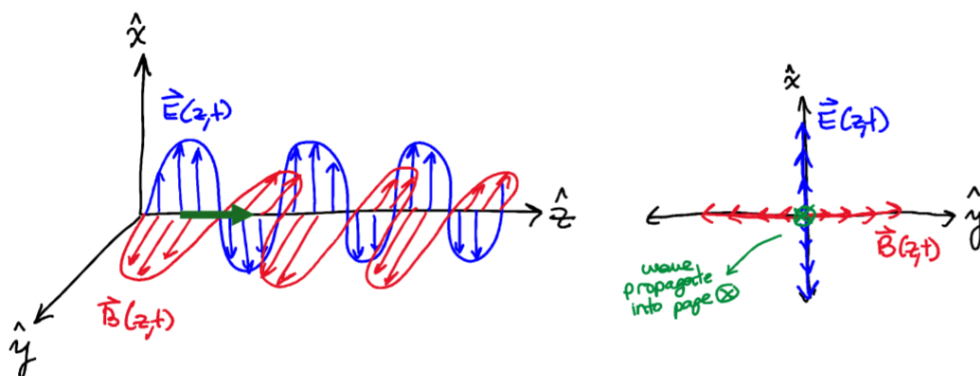


Note: Although light consists of both electric and magnetic fields, we will primarily focus on the electric field when modeling light, since the magnetic field can be readily derived from it using Faraday's law. We will also focus on the electric field when analyzing polarization, since the direction of the electric field determines the polarization of light.

Case $\hat{z}$

For the case $\hat{z}$, we take the direction of propagation to be along the $+z$ axis. At any fixed position z , the electric field $\vec{E}$ has a uniform magnitude and direction across the entire xy plane. As with the 1D plane wave, the *local view* at a fixed point along the z axis shows the electric field oscillating in time as the light wave passes through, with the electric field remaining transverse to the direction of propagation.

Note: Don't forget that the magnetic field $\vec{B}$ is also oscillating in phase with and is perpendicular to both $\vec{E}$ and the direction of propagation.



Therefore, a solution for the electric field of a plane wave in case $\hat{z}$ is:

$$\vec{E} = \underbrace{(E_{0x} e^{i\phi_x} \hat{x} + E_{0y} e^{i\phi_y} \hat{y})}_{\text{complex amplitude vector}} e^{i(kz - \omega t)}$$

From the equation, we can see that the magnitudes of the electric field in the x and y directions need not be the same, and that there is no z component of the electric field, since the wave is transverse. Like the 1D plane wave, the space and time dependence of this 3D plane wave is contained in the phase $(kz - \omega t)$.

The amplitude of the electric field is given by

$$|\vec{E}| = \sqrt{|E_{0x}|^2 + |E_{0y}|^2}$$

Note: The phases ϕ_x and ϕ_y of the complex amplitude vector do not affect the magnitude of the electric field, but they do govern the polarization of the light wave. This is because of the dot product:

$$|\vec{E}|^2 = \vec{E} \cdot \vec{E}^*, \quad \vec{E}^* = (\text{complex conjugate of } \vec{E})$$

$$\vec{E} \cdot \vec{E}^* = (E_{0x} e^{i\phi_x} \hat{x} + E_{0y} e^{i\phi_y} \hat{y}) \cdot (E_{0x} e^{-i\phi_x} \hat{x} + E_{0y} e^{-i\phi_y} \hat{y}) = |E_{0x}|^2 + |E_{0y}|^2$$

The exponentials multiply to 1 (e^0), and $\hat{x} \cdot \hat{x} = \hat{y} \cdot \hat{y} = 1$, while $\hat{x} \cdot \hat{y} = \hat{y} \cdot \hat{x} = 0$, so the cross terms cancel out.

Note: The xy components for both the electric and magnetic fields have the same harmonic properties of k and ω (wavelength and frequency), since they are both derived from the same phase ($kz - \omega t$).

Finally, note that we have switched from sinusoidal functions to complex exponentials. This is because complex exponentials are easier to work with mathematically, and if we want to get back to the real-valued (physically measurable) electric field, we can simply take the real part of the complex expression (using Euler's formula).

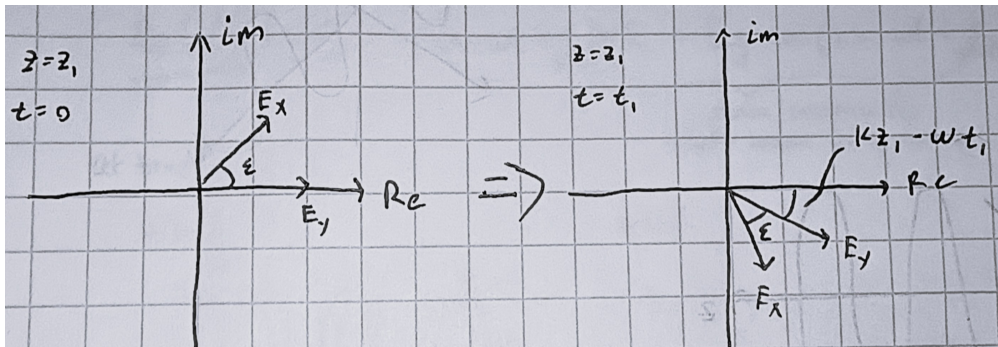
$$\vec{E}_{\text{real}} = \Re\{\vec{E}\} = \Re\{(E_{0x} e^{i\phi_x} \hat{x} + E_{0y} e^{i\phi_y} \hat{y}) e^{i(kz - \omega t)}\}$$

$$\vec{E}_{\text{real}} = E_{0x} \cos(kz - \omega t + \phi_x) \hat{x} + E_{0y} \cos(kz - \omega t + \phi_y) \hat{y}$$

Polarization of Light

Polarization has been mentioned a few times already, but what exactly is it? Simply put, polarization describes the path traced out by the tip of the electric field vector $\vec{E}$ in the xy plane at a fixed z as the light wave passes through it. How can we visualize this? With *parametric equations*.

First, to better understand the phase difference between the x and y components of the electric field, we draw a phasor diagram in the complex plane:



Where

$$E_x = E_{0x} e^{i(\phi_x + kz_1 - \omega t)}$$

$$E_y = E_{0y} e^{i(\phi_y + kz_1 - \omega t)}$$

$$\boxed{\varepsilon = \phi_x - \phi_y \quad (\text{phase difference})}$$

Since $z = z_1$ is fixed, kz_1 becomes a constant phase offset, so the rotation of the phasors is driven entirely by $-\omega t$ (CW, since it is negative). The phase $kz_1 - \omega t$ is the same for both E_x and E_y , so the relative phase difference ε between the two components remains constant as the light wave propagates. Also note that while the magnitudes of E_x and E_y are constant in the phasor diagram, their *real* magnitudes are given by the projections of the phasors onto the real axis, which oscillate in time (allowing elliptical polarization, where the magnitude of $\vec{E}$ varies in time).

General Polarization

The most general case of polarization is elliptical polarization, where linear and circular polarization are just degenerate ellipses. To see this, we will rewrite the *real* electric field components in terms of the phase difference ε :

$$\begin{aligned} \vec{E}_{\text{real}} &= E_{0x} \cos(kz_1 - \omega t + \phi_x) \hat{x} + E_{0y} \cos(kz_1 - \omega t + \phi_y) \hat{y} \\ x(t) &= E_{0x} \cos(\omega t - \varepsilon) \\ y(t) &= E_{0y} \cos(\omega t) \end{aligned}$$

Note: kz_1 is constant, so it cancels out when we take the difference of the two phases, but we keep ωt in the argument since it is the time-varying part. Because we define ε to be positive, it appears in the x component (which is why E_x leads E_y in the phasor diagram). The argument of the cosine is $\varepsilon - \omega t$, but since cosine is an even function, we can equivalently write it as $\omega t - \varepsilon$.

We now deparametrize the equations to obtain a single equation in x and y that describes the path traced out by $\vec{E}$ in the xy plane. First expanding the cosine using sum and difference:

$$\begin{aligned} x(t) &= E_{0x} (\cos(\omega t) \cos(\varepsilon) + \sin(\omega t) \sin(\varepsilon)) \\ y(t) &= E_{0y} \cos(\omega t) \implies \cos(\omega t) = \frac{y}{E_{0y}} \end{aligned}$$

Substituting $\cos(\omega t)$ into the x equation:

$$x(t) = E_{0x} \left(\frac{y}{E_{0y}} \cos(\varepsilon) + \sin(\omega t) \sin(\varepsilon) \right)$$

Solving for $\sin(\omega t)$:

$$\sin(\omega t) = \frac{1}{\sin(\varepsilon)} \left(\frac{x}{E_{0x}} - \frac{y}{E_{0y}} \cos(\varepsilon) \right)$$

Plugging $\sin(\omega t)$ and $\cos(\omega t)$ into the Pythagorean identity $\cos^2(\omega t) + \sin^2(\omega t) = 1$:

$$\left(\frac{y}{E_{0y}} \right)^2 + \left(\frac{1}{\sin(\varepsilon)} \left(\frac{x}{E_{0x}} - \frac{y}{E_{0y}} \cos(\varepsilon) \right) \right)^2 = 1$$

Simplify:

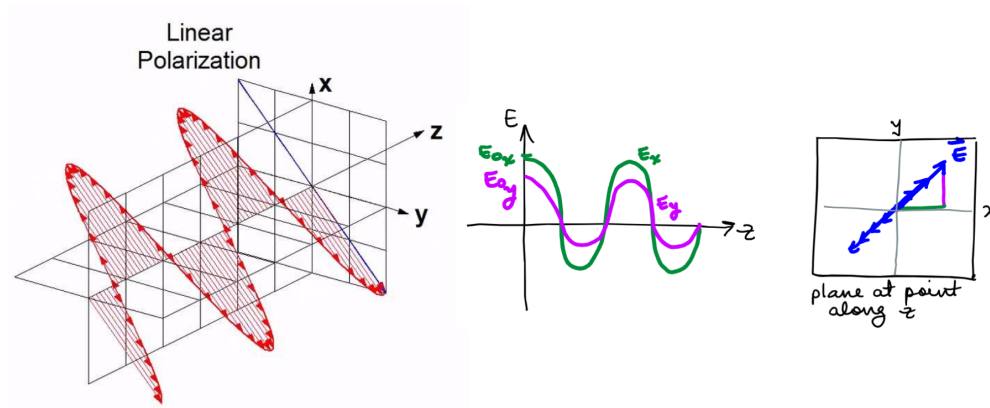
$$\frac{x^2}{E_{0x}^2} + \frac{y^2}{E_{0y}^2} - 2\frac{xy}{E_{0x}E_{0y}}\cos(\varepsilon) = \sin^2(\varepsilon)$$

Note: This is the form of an ellipse centered at the origin ($ax^2 + bxy + cy^2 = d$)

Therefore, for any ε that doesn't result in a degenerate case, the polarization of light is elliptical. Now we will list some cases so we don't have to keep referring back to the general equation:

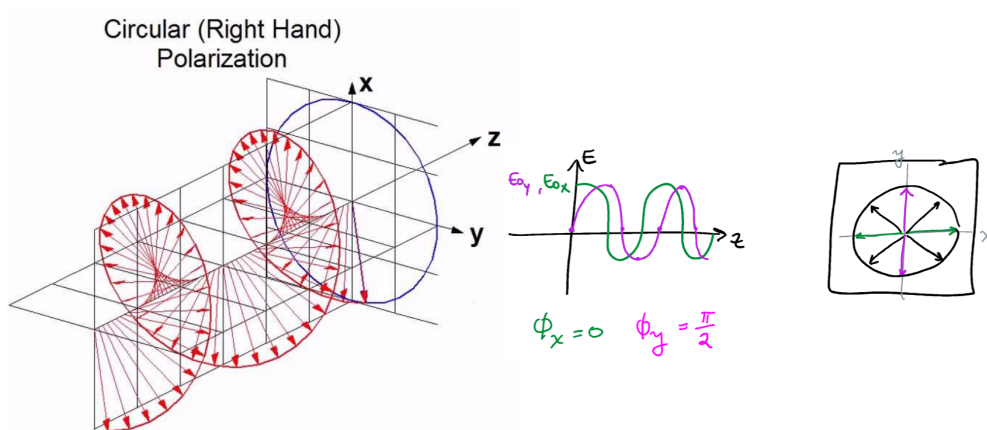
- **Linear Polarization:** E_x and E_y are in phase and the ratio of their magnitudes is constant (slope of the line).

$$\varepsilon = n\pi, \quad n = (0, 1, 2, \dots)$$



- **Circular Polarization:** E_x and E_y have equal magnitudes and are $\pi/2$ out of phase.

$$E_{0x} = E_{0y}, \quad \text{and} \quad \varepsilon = (2n + 1)\frac{\pi}{2}, \quad n = (0, 1, 2, \dots)$$

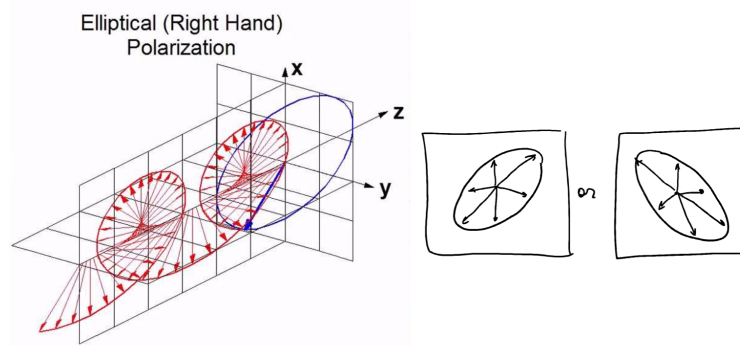


- **Elliptical Polarization:** All other cases

$$E_{0x} \neq E_{0y}, \quad \text{and} \quad \varepsilon \neq n\pi \quad (\text{otherwise it will be linear})$$

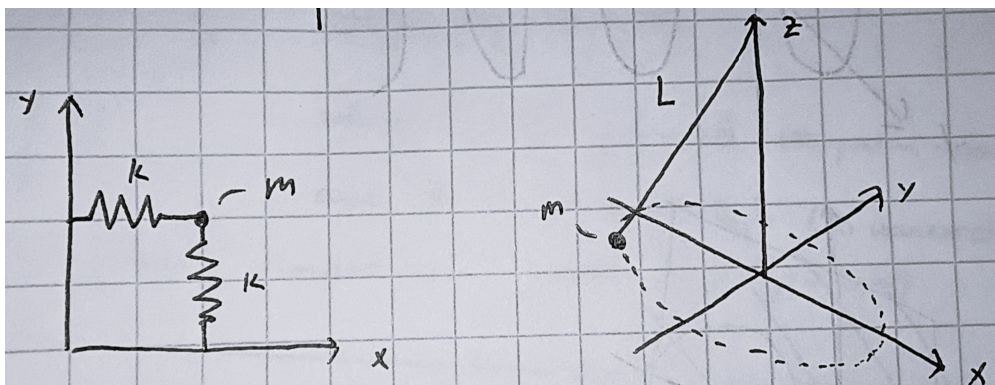
OR

$$E_{0x} = E_{0y}, \quad \text{and} \quad \varepsilon \neq (2n+1)\frac{\pi}{2} \quad (\text{otherwise it will be circular})$$



Why Ellipses?

Fundamentally, the electric and magnetic fields behave like oscillators, so the path traced out by the electric field vector must be periodic and closed. The only conic section that is both periodic and closed is an ellipse, with circles and lines appearing as degenerate cases. Another way to visualize this is to think of E_x and E_y as two simple harmonic oscillators (SHOs) that are coupled through a common driving frequency, analogous to two springs with equal spring constant k attached to the same mass. The motion of the mass is then a superposition of the two SHOs, and the resulting path traced out by the mass is an ellipse. Feynman also introduces the picture of a string-mass pendulum, where the mass can swing in any direction in the xy plane, and the path traced out by the mass is likewise an ellipse. This intuition is useful because it emphasizes that the driving phase ωt is the same for both the x and y components, and that the phase difference, and thus the resulting trace, depends only on the initial orientation of the motion.



General Plane Wave Solution

We would like to generalize the case $\hat{z}$ solution

$$\vec{E} = (E_{0_x} e^{i\phi_x} \hat{x} + E_{0_y} e^{i\phi_y} \hat{y}) e^{i(kz - \omega t)}$$

to describe a plane wave propagating in an arbitrary direction. To do this, we can simply replace the kz term in the phase with $\vec{k} \cdot \vec{r}$, where $\vec{k}$ is the **wavevector** that points in the direction of propagation and has magnitude k . This allows us to describe plane waves propagating in any direction, not just along the z axis.

$$\begin{aligned} \vec{k} &= k_x \hat{x} + k_y \hat{y} + k_z \hat{z} \\ |\vec{k}| &= \sqrt{k_x^2 + k_y^2 + k_z^2} = k, \quad k = \frac{2\pi}{\lambda} \end{aligned}$$

and $\vec{r}$ is the position vector from to origin to any point in space:

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

Therefore, the general plane wave solution is:

$$\boxed{\vec{E} = (E_{0_x} e^{i\phi_x} \hat{x} + E_{0_y} e^{i\phi_y} \hat{y}) e^{i(\vec{k} \cdot \vec{r} - \omega t)}}$$

This general form of the plane wave solution is intuitive because it is consistent with our understanding that $\vec{E}$ depends only on spatial variations along the direction of propagation. The dot product $\vec{k} \cdot \vec{r}$ extracts the component of $\vec{r}$ in the direction of $\vec{k}$, which for case $\hat{z}$ reduces to variations along the z axis. At the same time, $\vec{E}$ does not depend on spatial variations perpendicular to the direction of propagation (it is a transverse wave). For any $\vec{r}$ perpendicular to $\vec{k}$, we have $\vec{k} \cdot \vec{r} = 0$, so in case $\hat{z}$ there are no variations in the x and y directions.

Useful Properties of the General Plane Wave Form

Time derivative

$$\frac{\partial \vec{E}}{\partial t} = -i\omega \vec{E}, \quad \frac{\partial^2 \vec{E}}{\partial t^2} = -\omega^2 \vec{E}$$

Spatial derivative

$$\begin{aligned} \vec{\nabla} \cdot \vec{E} &= i\vec{k} \cdot \vec{E}, \quad \nabla^2 \vec{E} = -k^2 \vec{E} \\ \vec{\nabla} \times \vec{E} &= i\vec{k} \times \vec{E} \end{aligned}$$

Note: These properties follow from the fact that the only spatial and temporal dependence of $\vec{E}$ is contained in the phase $(\vec{k} \cdot \vec{r} - \omega t)$. When taking derivatives, the complex amplitude vector can therefore be treated as a constant, and the chain rule applies directly to the exponential term.

Also since we can find the direction of the magnetic field $\vec{B}$ using Faraday's law ($\vec{\nabla} \times \vec{E} = i\hat{k} \times \vec{E}$) and the relationship between the magnitudes of $\vec{E}$ and $\vec{B}$ ($|\vec{E}| = c|\vec{B}|$), we can also write:

$$\boxed{\vec{B} = \frac{1}{c} \hat{k} \times \vec{E}}$$

Constraints on Plane Wave Solutions from Maxwell's Equations

For **case $\hat{z}$** , we have the general plane wave solution of the form

$$\vec{E} = E_x \hat{x} + E_y \hat{y} + E_z \hat{z}$$

$$\vec{B} = B_x \hat{x} + B_y \hat{y} + B_z \hat{z}$$

Where E_x , E_y , E_z , B_x , B_y , and B_z are complex functions of the form:

$$E_x = E_{0x} e^{i(kz - \omega t + \phi_{x_E})}$$

Using the properties of time and spatial derivatives of the general plane wave form, we can apply Maxwell's equations to derive constraints on the components of $\vec{E}$ and $\vec{B}$.

(1) Applying Gauss's law for electricity:

$$\vec{\nabla} \cdot \vec{E} = 0$$

$$\left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \right) = 0$$

Since E_x and E_y are only dependent on z and t , their spatial derivatives with respect to x and y are zero.

$$\frac{\partial E_x}{\partial x} = \frac{\partial E_y}{\partial y} = 0$$

Thus,

$$\begin{aligned} \frac{\partial E_z}{\partial z} &= \frac{\partial}{\partial z} (E_{0z} e^{i(kz - \omega t + \phi_{z_E})}) = 0 \\ \implies ikE_{0z} e^{i(kz - \omega t + \phi_{z_E})} &= 0 \end{aligned}$$

Since the exponential term is never zero, $\boxed{E_{0z} \text{ must be zero}}$.

This matches our expectation that the electric field of a plane wave must be transverse to the direction of propagation, so there can be no z component of the electric field for case $\hat{z}$.

(2) Applying Gauss's law for magnetism:

$$\vec{\nabla} \cdot \vec{B} = 0$$

Similarly to (1), we can see that B_{0z} must also be zero. Also note that the lack of an x and y dependence in both $\vec{E}$ and $\vec{B}$ shows that the fields are *uniform* across the entire xy plane, which is consistent with the plane wave approximation.

(3) Applying Faraday's law: Setting E_{0z} and B_{0z} to zero as found in (1) and (2).

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\begin{aligned}
\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & 0 \end{vmatrix} &= -\frac{\partial}{\partial z} E_y \hat{x} + \frac{\partial}{\partial z} E_x \hat{y} + \underbrace{\left(\frac{\partial}{\partial x} E_y - \frac{\partial}{\partial y} E_x \right)}_0 \hat{z} \\
&= -\frac{\partial}{\partial z} (E_{0y} e^{i(kz-\omega t+\phi_{yE})}) \hat{x} + \frac{\partial}{\partial z} (E_{0x} e^{i(kz-\omega t+\phi_{xE})}) \hat{y} \\
&= -ikE_{0y} e^{i(kz-\omega t+\phi_{yE})} \hat{x} + ikE_{0x} e^{i(kz-\omega t+\phi_{xE})} \hat{y} \\
&= ik e^{i(kz-\omega t)} (-E_{0y} e^{i\phi_{yE}} \hat{x} + E_{0x} e^{i\phi_{xE}} \hat{y})
\end{aligned}$$

Now solving for $-\frac{\partial \vec{B}}{\partial t}$.

$$\begin{aligned}
-\frac{\partial}{\partial t} (B_x \hat{x} + B_y \hat{y}) &= -\left[\frac{\partial}{\partial t} (B_{0x} e^{i(kz-\omega t+\phi_{xB})}) \hat{x} + \frac{\partial}{\partial t} (B_{0y} e^{i(kz-\omega t+\phi_{yB})}) \hat{y} \right] \\
&= -[-i\omega B_{0x} e^{i(kz-\omega t+\phi_{xB})} \hat{x} - i\omega B_{0y} e^{i(kz-\omega t+\phi_{yB})} \hat{y}] \\
&= i\omega e^{i(kz-\omega t)} (B_{0x} e^{i\phi_{xB}} \hat{x} + B_{0y} e^{i\phi_{yB}} \hat{y})
\end{aligned}$$

Setting them equal to each other gives:

$$k (-E_{0y} e^{i\phi_{yE}} \hat{x} + E_{0x} e^{i\phi_{xE}} \hat{y}) = \omega (B_{0x} e^{i\phi_{xB}} \hat{x} + B_{0y} e^{i\phi_{yB}} \hat{y})$$

Looking at the x components, the magnitudes and phases must match:

$$\begin{aligned}
-kE_{0y} e^{i\phi_{yE}} &= \omega B_{0x} e^{i\phi_{xB}} \\
\implies E_{0y} e^{i\pi} e^{i\phi_{yE}} &= \frac{\omega}{k} B_{0x} e^{i\phi_{xB}}
\end{aligned}$$

Thus,

$$\boxed{\frac{E_{0y}}{B_{0x}} = \frac{\omega}{k} = c, \quad \phi_{yE} + \pi = \phi_{xB}}$$

Looking at the y components, the magnitudes and phases must match:

$$\begin{aligned}
kE_{0x} e^{i\phi_{xE}} &= \omega B_{0y} e^{i\phi_{yB}} \\
\implies E_{0x} e^{i\phi_{xE}} &= \frac{\omega}{k} B_{0y} e^{i\phi_{yB}}
\end{aligned}$$

Thus,

$$\boxed{\frac{E_{0x}}{B_{0y}} = \frac{\omega}{k} = c, \quad \phi_{xE} = \phi_{yB}}$$

(4) Applying Ampere-Maxwell law: Setting E_{0z} and B_{0z} to zero as found in (1) and (2).

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$$

We will see the same relationships found in (3) emerge, physics is consistent!

$$\begin{aligned}
\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & B_y & 0 \end{vmatrix} &= -\frac{\partial}{\partial z} B_y \hat{x} + \frac{\partial}{\partial z} B_x \hat{y} + \underbrace{\left(\frac{\partial}{\partial x} B_y - \frac{\partial}{\partial y} B_x \right)}_0 \hat{z} \\
&= -\frac{\partial}{\partial z} (B_{0y} e^{i(kx - \omega t + \phi_{yB})}) \hat{x} + \frac{\partial}{\partial z} (B_{0x} e^{i(kx - \omega t + \phi_{xB})}) \hat{y} \\
&= i k e^{i(kx - \omega t)} [-B_{0y} e^{i\phi_{yB}} \hat{x} + B_{0x} e^{i\phi_{xB}} \hat{y}]
\end{aligned}$$

Now solving for $\frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$.

$$\begin{aligned}
\frac{1}{c^2} \frac{\partial}{\partial t} (E_x \hat{x} + E_y \hat{y}) &= \frac{1}{c^2} \left[\frac{\partial}{\partial t} (E_{0x} e^{i(kx - \omega t + \phi_{xE})}) \hat{x} + \frac{\partial}{\partial t} (E_{0y} e^{i(kx - \omega t + \phi_{yE})}) \hat{y} \right] \\
&= \frac{1}{c^2} [-i\omega E_{0x} e^{i(kx - \omega t + \phi_{xE})} \hat{x} - i\omega E_{0y} e^{i(kx - \omega t + \phi_{yE})} \hat{y}] \\
&= -\frac{1}{c^2} i\omega e^{i(kx - \omega t)} [E_{0x} e^{i\phi_{xE}} \hat{x} + E_{0y} e^{i\phi_{yE}} \hat{y}]
\end{aligned}$$

$\hat{x}$ components:

$$\begin{aligned}
-k B_{0y} e^{i\phi_{yB}} &= -\frac{1}{c^2} \omega E_{0x} e^{i\phi_{xE}} \\
\Rightarrow B_{0y} e^{i\phi_{yB}} &= \frac{1}{c^2} \frac{\omega}{k} E_{0x} e^{i\phi_{xE}} \quad \Rightarrow \quad \boxed{B_{0y} = \frac{1}{c} E_{0x}, \quad \phi_{yB} = \phi_{xE}}
\end{aligned}$$

$\hat{y}$ components:

$$\begin{aligned}
k B_{0x} e^{i\phi_{xB}} &= -\frac{1}{c^2} \omega E_{0y} e^{i\phi_{yE}} \\
\Rightarrow B_{0x} e^{i\phi_{xB}} &= \frac{1}{c^2} \frac{\omega}{k} E_{0y} e^{i\pi} e^{i\phi_{yE}} \quad \Rightarrow \quad \boxed{B_{0x} = \frac{1}{c} E_{0y}, \quad \phi_{xB} = \phi_{yE} + \pi}
\end{aligned}$$

A *slightly* better method

Regrettably, while this previous method is rigorous, it is also very tedious and didn't use our newfound properties of general plane wave derivatives! Below is a (quick, I hope) method from Professor Corn:

We saw from before that when applying Faraday's law, we get

$$\begin{aligned}
\vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\
-\frac{\partial E_y}{\partial z} \hat{x} + \frac{\partial E_x}{\partial z} \hat{y} &= -\frac{\partial B_x}{\partial t} \hat{x} - \frac{\partial B_y}{\partial t} \hat{y}
\end{aligned}$$

Setting the terms equal:

$$\Rightarrow \frac{\partial E_y}{\partial z} = \frac{\partial B_x}{\partial t} \quad \text{and} \quad \frac{\partial E_x}{\partial z} = -\frac{\partial B_y}{\partial t}$$

From the general plane wave solution:

$$\begin{aligned}\frac{\partial E_y}{\partial z} &= \frac{\partial}{\partial z} [E_{0y} e^{i\phi_{yE}} e^{i(kz-\omega t)}] = ikE_y \quad \text{and} \quad \frac{\partial E_x}{\partial z} = ikE_x \\ \frac{\partial B_x}{\partial t} &= \frac{\partial}{\partial t} [B_{0x} e^{i\phi_{xB}} e^{i(kz-\omega t)}] = -i\omega B_x \quad \text{and} \quad \frac{\partial B_y}{\partial t} = -i\omega B_y \\ \implies ikE_y &= -i\omega B_x \implies ikE_x = -(-i\omega B_y) \\ \frac{E_y}{B_x} &= \frac{-\omega}{k} = -c \quad \frac{E_x}{B_y} = \frac{\omega}{k} = c\end{aligned}$$

Plugging in for E_x , E_y , B_x , and B_y :

$$\frac{E_y}{B_x} = \frac{E_{0y} e^{i\phi_{yE}} e^{i(kz-\omega t)}}{B_{0x} e^{i\phi_{xB}} e^{i(kz-\omega t)}} = \frac{E_{0y}}{B_{0x}} e^{i(\phi_{yE} - \phi_{xB})} = -c, \quad -c = ce^{i\pi}$$

Equating the magnitudes and phases once again gives:

$$\frac{E_{0y}}{B_{0x}} = c \quad \text{and} \quad \phi_{yE} - \phi_{xB} = \pi$$

Similarly for the x component:

$$\frac{E_x}{B_y} = \frac{E_{0x} e^{i\phi_{xE}} e^{i(kz-\omega t)}}{B_{0y} e^{i\phi_{yB}} e^{i(kz-\omega t)}} = \frac{E_{0x}}{B_{0y}} e^{i(\phi_{xE} - \phi_{yB})} = c$$

Equating the magnitudes and phases once again gives:

$$\frac{E_{0x}}{B_{0y}} = c \quad \text{and} \quad \phi_{xE} - \phi_{yB} = 0$$

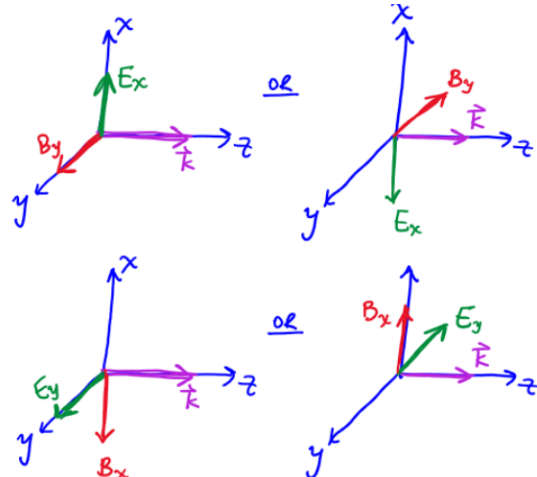
Note: If we didn't add in a phase difference of π then we would see that $E_{0y}/B_{0x} = -\omega/k$. In either case, the right-hand rule requires E_{0y} and B_{0x} to have opposite signs since $\vec{E} \times \vec{B}$ always points in the $\vec{k}$ direction. This can be expressed either by introducing a phase difference of π or by keeping the negative sign in the ratio.

Assuming $\vec{k}$ is in $+\hat{z}$, then either E_{0x} and B_{0y} are both positive or both negative!

$$\implies \frac{E_{0x}}{B_{0y}} = c$$

Assuming $\vec{k}$ is in $+\hat{z}$, then E_{0y} and B_{0x} must have opposite signs!

$$\implies \frac{E_{0y}}{B_{0x}} = -c$$



Energy Transported by EM Waves

Recall that energy is stored in the electric and magnetic fields, with energy densities given by

$$u_E = \frac{1}{2}\epsilon_0|\vec{E}|^2 \quad (\text{energy density of electric field})$$

$$u_B = \frac{1}{2\mu_0}|\vec{B}|^2 \quad (\text{energy density of magnetic field})$$

Note: As mentioned before, since we are now dealing with physically measurable quantities of the $\vec{E}$ and $\vec{B}$ fields, we need to use the *real* components of the fields.

Additionally, we are ignoring phase differences for now to simplify calculations, so $\phi_x = \phi_y = \phi_z = 0$ for the electric field (this simplification also applies to the magnetic field, but we'll focus on just the electric field for now).

$$\vec{E} = \Re\left\{\vec{E}_{\text{complex}}\right\} = \Re\left\{(E_{0x}\hat{x} + E_{0y}\hat{y} + E_{0z}\hat{z})e^{i(\vec{k}\cdot\vec{r}-\omega t)}\right\}$$

$$\vec{E} = E_{0x}\cos(\vec{k}\cdot\vec{r}-\omega t)\hat{x} + E_{0y}\cos(\vec{k}\cdot\vec{r}-\omega t)\hat{y} + E_{0z}\cos(\vec{k}\cdot\vec{r}-\omega t)\hat{z}$$

Now the magnitude of the electric field is given by the dot product:

$$|\vec{E}|^2 = \vec{E} \cdot \vec{E} = |\vec{E}_0|^2 \cos^2(\vec{k}\cdot\vec{r}-\omega t), \quad \text{where} \quad \vec{E}_0 = E_{0x}\hat{x} + E_{0y}\hat{y} + E_{0z}\hat{z}$$

For EM Waves

- Since the magnitudes of the electric and magnetic fields are related by $|\vec{E}| = c|\vec{B}|$, the energy densities of the electric and magnetic fields are also equal ($c = \frac{1}{\sqrt{\epsilon_0\mu_0}}$):

$$\boxed{u_E = u_B}$$

- The **total energy density** u is then the sum of the electric and magnetic energy densities:

$$\boxed{u = u_E + u_B = 2u_E = \epsilon_0|\vec{E}|^2}$$

Poynting Vector $\vec{S}$ We can derive the Poynting vector $\vec{S}$, which describes the flow of electromagnetic energy carried by the $\vec{E}$ and $\vec{B}$ fields in the direction of $\vec{k}$. Its units are power per unit area, W/m². We can derive $\vec{S}$ by applying the *continuity equation* (local conservation of energy) to the energy density u . We should expect to see that:

$$\frac{\partial u}{\partial t} + \vec{\nabla} \cdot \vec{S} = 0,$$

Similar to how we relate the change in charge density to the divergence of current density in the continuity equation for charge, here the change in energy density is related to the divergence of the Poynting vector which describes the rate at which energy flows through an area.

Starting from $u = \frac{1}{2}\epsilon_0 \vec{E} \cdot \vec{E} + \frac{1}{2\mu_0} \vec{B} \cdot \vec{B}$, we want to show that $\frac{\partial u}{\partial t} = -\vec{\nabla} \cdot \vec{S}$:

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial t} \left[\frac{1}{2}\epsilon_0 \vec{E} \cdot \vec{E} + \frac{1}{2\mu_0} \vec{B} \cdot \vec{B} \right] = \frac{1}{2}\epsilon_0 \frac{\partial}{\partial t} (\vec{E} \cdot \vec{E}) + \frac{1}{2\mu_0} \frac{\partial}{\partial t} (\vec{B} \cdot \vec{B})$$

Applying the product rule to the dot products:

$$\begin{aligned} \frac{\partial}{\partial t} (\vec{E} \cdot \vec{E}) &= 2\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \quad \text{and} \quad \frac{\partial}{\partial t} (\vec{B} \cdot \vec{B}) = 2\vec{B} \cdot \frac{\partial \vec{B}}{\partial t} \\ \implies \frac{\partial u}{\partial t} &= \epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} + \frac{1}{\mu_0} \vec{B} \cdot \frac{\partial \vec{B}}{\partial t} \end{aligned}$$

From Faraday's Law:

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \implies \frac{\partial \vec{B}}{\partial t} = -\vec{\nabla} \times \vec{E}$$

From Ampere-Maxwell Law:

$$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \implies \frac{\partial \vec{E}}{\partial t} = \frac{1}{\mu_0 \epsilon_0} \vec{\nabla} \times \vec{B}$$

Substituting these into the expression for $\frac{\partial u}{\partial t}$:

$$\begin{aligned} \frac{\partial u}{\partial t} &= \epsilon_0 \vec{E} \cdot \left(\frac{1}{\mu_0 \epsilon_0} \vec{\nabla} \times \vec{B} \right) + \frac{1}{\mu_0} \vec{B} \cdot (-\vec{\nabla} \times \vec{E}) \\ &= \frac{1}{\mu_0} \left[\vec{E} \cdot (\vec{\nabla} \times \vec{B}) - \vec{B} \cdot (\vec{\nabla} \times \vec{E}) \right] \\ &= \frac{1}{\mu_0} \left[-\vec{\nabla} \cdot (\vec{E} \times \vec{B}) \right] \end{aligned}$$

Note: The last step uses the vector identity $\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})$.

Therefore, the Poynting vector is

$$\boxed{\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}}$$

It is also equivalent to

$$\begin{aligned} \vec{S} &= \frac{1}{\mu_0} \vec{E} \times \vec{B} = \frac{1}{\mu_0} |\vec{E}| \underbrace{|\vec{B}|}_{\frac{1}{c}|\vec{E}|} \hat{k} = \frac{1}{\mu_0} |\vec{E}|^2 \underbrace{\hat{k}}_{\epsilon_0 c} \\ \implies \boxed{\vec{S} = c u \hat{k}} \end{aligned}$$

Intensity

Intensity is defined as the time-averaged magnitude of the Poynting vector, which describes the *average* power per unit area carried by the electromagnetic wave.

$$\langle \vec{S} \rangle = c \langle u \rangle \hat{k}, \quad \langle \quad \rangle : \text{time average}$$

Where

$$\langle u \rangle = \varepsilon_0 \langle |\vec{E}|^2 \rangle = \varepsilon_0 E_0^2 \langle \cos^2(\vec{k} \cdot \vec{r} - \omega t) \rangle = \frac{1}{2} \varepsilon_0 E_0^2$$

The time average of $\cos^2$ is $1/2$ because the function oscillates between 0 and 1, and over a full period, it spends equal time at all values in that range. Therefore, the average value of $\cos^2$ over one period is $1/2$.

Thus, the intensity of an electromagnetic wave is given by

$$I = |\langle \vec{S} \rangle| = c \langle u \rangle = \frac{1}{2} \varepsilon_0 E_0^2 c$$

Note: A higher light intensity corresponds to a larger electric-field amplitude, meaning the wave transports energy at a greater rate per unit area. This aligns with the physical interpretation of intensity as a measure of brightness: brighter light delivers more energy to a surface per unit time.

When dealing with electromagnetic waves, taking the time average is important because the fields oscillate rapidly, and we typically observe them on a macroscopic scale where the average behavior is more meaningful than the instantaneous values.

Unit 2: Radiation and Scattering

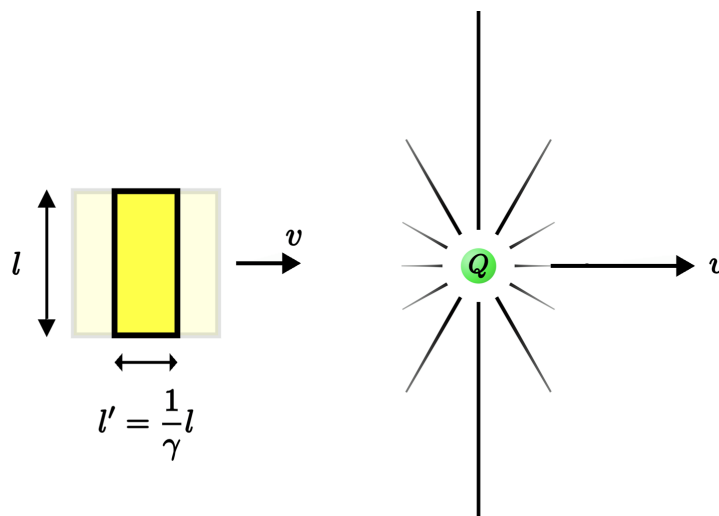
The goal of this unit is to model how electromagnetic waves are generated by an accelerated charge (via radiation and scattering) using *classical physics*. Along the way, we will make some approximations and assumptions that will yield a usable model but also introduce some limitations that we will later address with quantum mechanics.

Thompson Radiation

In the Thompson model, we are considering the electromagnetic radiation emitted by a *free* charged particle previously travelling at a constant velocity (or at rest) that is accelerated by a short *constant* acceleration, a "kick". This acceleration causes a kink in the electric field lines of the charge, which then propagates outward as an electromagnetic wave.

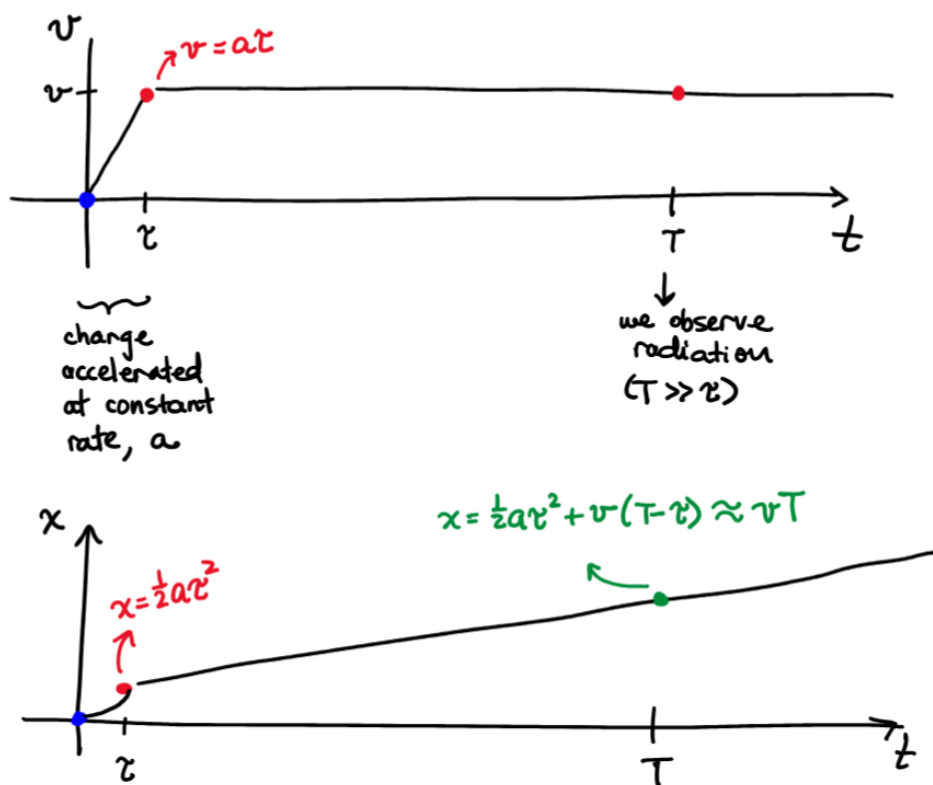
Before we begin, it is important to note the *scale* of the set-up we are working with. Perhaps the most important simplifying approximation we make is that the duration of the acceleration τ is very short compared to the time *after the acceleration* at which we observe the electric field, T ($\tau \ll T$). Additionally, τ is so short that the final constant velocity of the charge after the acceleration remains much less than the speed of light ($v \ll c$), so we can ignore relativistic effects. Importantly, this approximation lets us treat the electric field of a charge moving at constant velocity as essentially the same as that of a static charge, except that the field now points away from the *instantaneous position* of the charge rather than from a fixed point such as the origin.

For a charge moving relativistic velocity, something similar to length contraction occurs, where the electric field lines become squished along the direction of motion, and the field becomes stronger in the plane perpendicular to the motion and the lorentz factor γ appears. For our case, $v \ll c$, so $\gamma \approx 1$ and we can ignore these relativistic effects.



The Setup

1. A free charge is at rest at $x = 0$ for all time $t \leq 0$.
2. The charge is "kicked" and accelerated with a constant a for a short period of time τ in the $+x$ direction.
3. After the acceleration, the charge moves at a constant velocity $v = a\tau$ in the $+x$ direction forever ($t > \tau$).
4. We will observe the electric field lines far away from the charge and at a much later time T after the acceleration ($\tau \ll T$).



Relativistic Causality

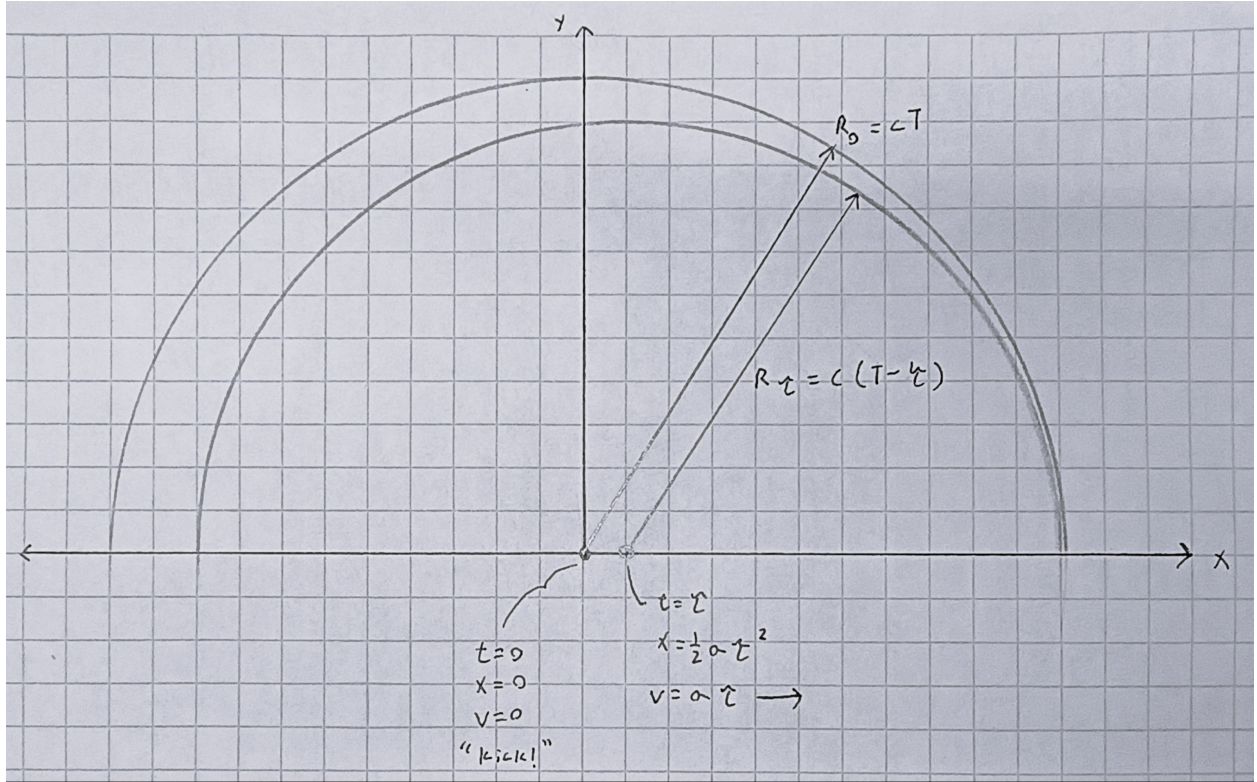
Relativistic causality states that no information can propagate faster than the speed of light. In this model, the information of interest is the beginning and end of the acceleration of the charge. These disturbances propagate outward at speed c .

The outer boundary, which represents the beginning of the acceleration, is a sphere of radius centered at the origin, corresponding to the physical position of the charge at $t = 0$.

$$R_0 = cT$$

Likewise, the inner boundary, which represents the end of the acceleration, is a sphere of radius centered at $x = \frac{1}{2}a\tau^2$ which is the physical position of the charge at $t = \tau$.

$$R_1 = c(T - \tau)$$



The region between these two radii contains the kink in the electric field, that is, the electromagnetic wave. This is why we say that an accelerated charge produces radiation.

Under the approximation $\tau \ll T$ and in the limit $\tau \rightarrow 0$, the displacement of the charge during the acceleration becomes negligible. Thus, the charge's position after the acceleration is approximately the same as its initial position, and the two spheres may be treated as approximately concentric. Additionally, following our previous approximation that the electric field of a charge moving at constant velocity is essentially the same as that of a static charge, we can say that the field lines entering and exiting this region are parallel, forming the same angles to the x axis.

Note: We can choose any Gaussian surface that encloses the charge. Since the electric field line is continuous, its magnitude remains the same along its entire path. However, choosing a sphere that lies entirely within region I is most convenient, because the electric flux there comes solely from the static charge at the origin due to relativistic causality. In region II, by contrast, the charge has moved and is no longer centered within the sphere, leading to a non-uniform electric flux distribution.

A more useful form for E_θ can be obtained by substituting $a = v/\tau$ and $R = cT$:

$$E_\theta = \frac{1}{4\pi\epsilon_0} \frac{qa \sin \theta}{c^2 R}$$

Therefore, the electric field kink is

$$\vec{E}_a = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} \hat{r} + \frac{1}{4\pi\epsilon_0} \frac{qa \sin \theta}{c^2 R} \hat{\theta}$$

Since the radial component E_r falls off as $1/R^2$ while the transverse component E_θ falls off as $1/R$, for large observation distances R , $E_r \ll E_\theta$ and the electric field is dominated by the transverse component. Therefore, for radiation, and also for quantifying the resulting electromagnetic wave, we can use just E_θ , the transverse component of the electric field.

This also makes intuitive sense in our model. Since $R = cT$, a large R corresponds to a large T , so the approximation $\tau \ll T$ becomes increasingly valid. In this limit, the region between regions I and II effectively shrinks to a spherical shell of negligible thickness, and the field lines become squished so that they are essentially tangential to the sphere, meaning they are purely transverse.

Finally, putting all the intuition and math together, we have the formula for the **Thompson radiation** generated from an accelerating charge:

$$\boxed{\vec{E}(\vec{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{qa \sin \theta}{c^2 |\vec{r}|} \hat{\theta} \quad (\text{Thompson radiation})}$$

Where:

- a is the acceleration of the charge.
- q is the charge of the particle.
- θ is the angle between the direction of acceleration and the direction of observation.
- c is the speed of light.

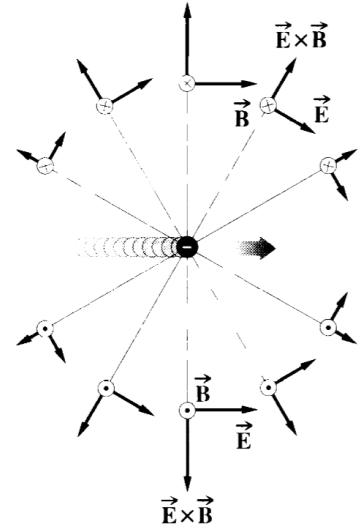
$\vec{r}$ is the position vector from the charge's initial position to the observation point, and is also the *direction of the electromagnetic wave propagation* $\vec{k}$. This is because the kink in the electric field lines spreads out radially outward, while the electric field itself is always tangential to $\hat{r}$.

Also, while time t doesn't explicitly appear in the formula physically, recall that the radius $|\vec{r}|$ is related to the time of observation T by $|\vec{r}| = cT$. Additionally, the acceleration a can also be a function of time, and the formula for Thompson radiation can be applied to each instantaneous value of a .

As a concluding thought, electromagnetic waves are generated by an accelerated charge fundamentally because of relativistic causality. The discrepancy between the travelling spheres of information about the beginning and the end of the acceleration simply can only be resolved by the electric field lines developing a kink in the region between the two spheres. From the arguments developed in the past four pages, this kink is not optional but required, and it is precisely this feature that constitutes the radiation field.

Note: Recall from the general plane wave solution that we can also define the magnetic field *induced* by the radiated electric field kink using

$$\vec{B}(\vec{r}, t) = \frac{1}{c} \hat{k} \times \vec{E}, \quad \hat{k} = \hat{r}$$



Poynting Flux

Using the Poynting flux vector, we can calculate the rate of energy radiated by the accelerating charge:

$$\vec{S}(\vec{r}, t) = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

$$|\vec{S}| = \varepsilon_0 c |\vec{E}|^2 = \frac{q^2 a^2 \sin^2 \theta}{16\pi^2 \varepsilon_0 r^2 c^3} \left[\frac{\text{W}}{\text{m}^2} \right]$$

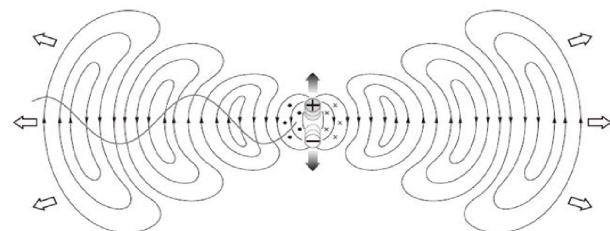
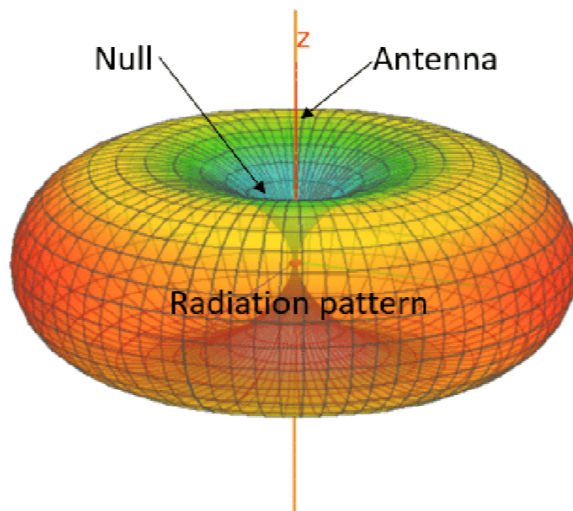
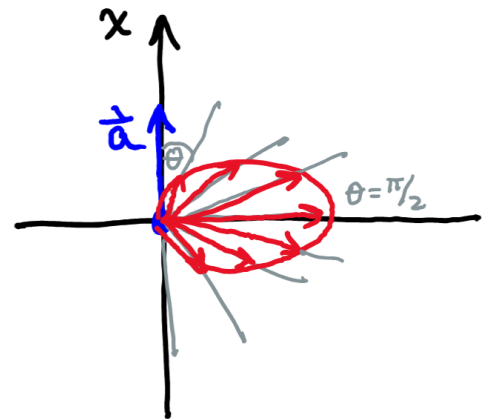
this is the power per unit area at a distance $r = |\vec{r}|$ from the charge and an angle θ from the direction of acceleration due to a short acceleration a .

What Does it Look Like?

The Poynting vector $\vec{S}$ points in the direction of electromagnetic wave propagation $\hat{r}$, so energy flows radially outward from the charge. Thus, the radiated power spreads away from the source in spherical wavefronts, like sound waves or ripples on a pond from an isotropic source.



Power depends on the angle θ , meaning that it is strongest at $\theta = \pi/2$ (perpendicular to the direction of acceleration) and weakest at $\theta = 0$ or π (along the direction of acceleration). This results in a doughnut (toroidal) radiation pattern, with the charge at the center and the hole of the doughnut along the axis of acceleration.



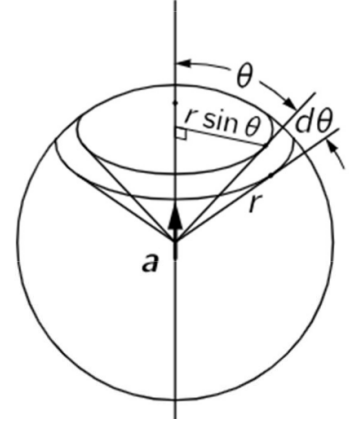
The $\vec{E}$ -field of an oscillating electric dipole.

Larmor Power

Larmor power is the total power radiated by an accelerating charge, which can be found by integrating the Poynting flux (power per area) over a spherical surface that contains the charge, thus capturing all the flux. The sphere is centered on the charge's initial position, and its radius r is arbitrary, since the total radiated power is independent of the distance from the charge.

Since $\vec{S}$ points in the $\hat{r}$ direction and is always normal to the sphere surface,

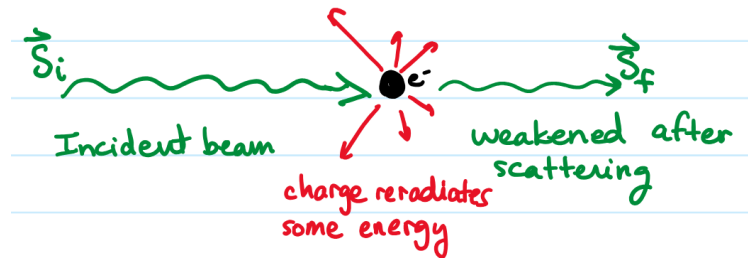
$$\begin{aligned}
 P &= \oint_S \vec{S} \cdot d\vec{A} = \int |\vec{S}| dA \\
 &= \int_0^{2\pi} \int_0^\pi |\vec{S}| r^2 \sin \theta d\theta d\phi \\
 &= \int_0^\pi |\vec{S}| 2\pi r^2 \sin \theta d\theta \\
 &= \int_0^\pi \frac{q^2 a^2 \sin^2 \theta}{16\pi^2 \epsilon_0 r^2 c^3} 2\pi r^2 \sin \theta d\theta \\
 &= \frac{q^2 a^2}{8\pi \epsilon_0 c^3} \underbrace{\int_0^\pi \sin^3 \theta d\theta}_{4/3}
 \end{aligned}$$



$$P_{\text{rad}} = \frac{q^2 a^2}{6\pi \epsilon_0 c^3} \quad (\text{Watts})$$

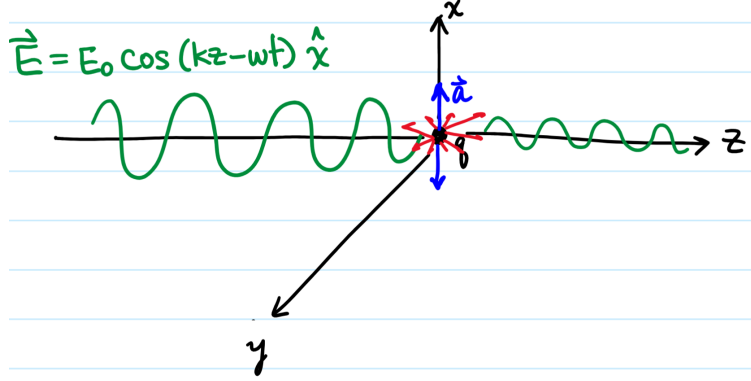
Scattering by Free Charges

Scattering occurs when an incoming electromagnetic wave interacts with a free charge (Thompson scattering) or a bound charge (Rayleigh scattering). The incoming wave causes the charge to accelerate, which will then *reradiate* the energy in all directions. Our goal is then to quantify the scattered radiation in terms of its power and how the initial wave is changed by the scattering process.



Thompson Scattering

We will begin with Thompson scattering which describes the scattering of an electromagnetic wave by a free charge. Consider the following incoming electromagnetic wave with the electric field:



A free charge at the origin ($z = 0$) will experience a force equal to:

$$\vec{F} = q\vec{E} = qE_0 \cos(\omega t) \hat{x}$$

$$\implies a = a_x = \frac{qE_0}{m} \cos(\omega t)$$

Plugging this into the formula for Larmor power gives the instantaneous power radiated by the accelerating charge:

$$P(t) = \frac{q^2 a^2}{6\pi\epsilon_0 c^3} = \frac{q^2}{6\pi\epsilon_0 c^3} \left(\frac{qE_0}{m} \cos(\omega t) \right)^2 = \frac{q^4 E_0^2}{6\pi\epsilon_0 m^2 c^3} \cos^2(\omega t) \quad (\text{Watts})$$

Taking the time average gives the average power radiated by the accelerating charge (which is also the power of the scattered wave):

$$\boxed{\langle P_{\text{scatt}}(t) \rangle = \frac{q^4 E_0^2}{12\pi\epsilon_0 m^2 c^3}}$$

Notice that $\langle P_{\text{scatt}}(t) \rangle$ is proportional to the square of the incident electric field amplitude E_0^2 , thus we can rewrite the power of the scattered wave to be related to the intensity $I_{\text{inc}} = |\langle \vec{S} \rangle_{\text{inc}}|$ of the incident wave:

$$|\langle \vec{S} \rangle_{\text{inc}}| = \frac{1}{2} \epsilon_0 E_0^2 c \implies E_0^2 = \frac{2|\langle \vec{S} \rangle_{\text{inc}}|}{\epsilon_0 c}$$

$$\langle P_{\text{scatt}}(t) \rangle = \frac{q^4}{12\pi\epsilon_0 m^2 c^3} \frac{2|\langle \vec{S} \rangle_{\text{inc}}|}{\epsilon_0 c} = \frac{8\pi}{3} \underbrace{\left(\frac{q^2}{4\pi\epsilon_0 m c^2} \right)^2}_{\text{area, m}^2} \underbrace{|\langle \vec{S} \rangle_{\text{inc}}|}_{\text{intensity W/m}^2}$$

Thompson Cross Section

We define the Thomson cross section σ_{Th} as the effective area that quantifies the likelihood of scattering by a free charge when an incident electromagnetic wave interacts with it. A useful mechanical analogy is to imagine shooting a stream of BB pellets at an 8 ball. If a BB hits the 8 ball, it bounces off, and the probability of this happening depends on the size of the 8 ball (a larger 8 ball increases the chance of being hit).

In this case, the Thomson cross section is not the *physical* cross-sectional area of the charge, since a charge doesn't exactly have a physical size. Instead, it is an *effective* area that captures the probability of scattering in the same intuitive way. Therefore,

$$\langle P_{\text{scatt}}(t) \rangle = \sigma_{\text{Th}} |\langle \vec{S} \rangle_{\text{inc}}|$$

Where

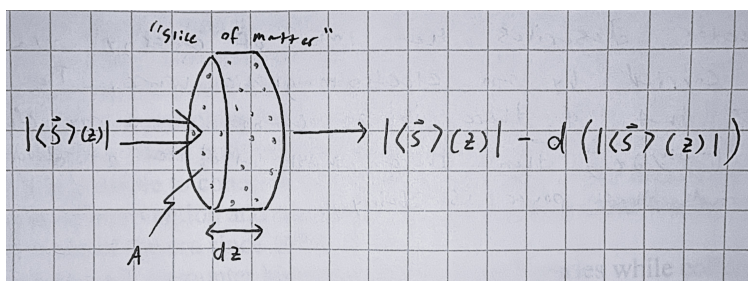
$$\sigma_{\text{Th}} = \frac{8\pi}{3} \left(\frac{q^2}{4\pi\epsilon_0 mc^2} \right)^2 \quad (\text{general cross-section})$$

$$\sigma_{\text{Th}} = \frac{8\pi}{3} r_e^2 \quad (\text{electron cross-section})$$

Passing Through a Material

So far we have quantified the power of the scattered wave from *a single free charge*. To find the *total* power scattered by a volume that contains many free charges, we can simply multiply the power scattered per charge by the number of charges in the volume.

Consider a slice of matter with thickness dz , cross-sectional area A , and number density of free charges N (the number of free charges per unit volume), where each free charge has a Thomson cross section σ_{Th} . We aim to derive an expression for the change in intensity $d(|\langle \vec{S} \rangle_{\text{inc}}|)$ due to scattering by the free charges within this volume. This scattering removes energy from the forward direction, causing a decrease in the intensity of the transmitted wave.



The power entering the volume is given by: $|\langle \vec{S} \rangle(z)| A$

similarly, the exiting power from the volume is given by: $|\langle \vec{S} \rangle(z)| A - d(|\langle \vec{S} \rangle(z)|) A$

Then we can define the total power lost by scattering in the volume as:

$$-A d(|\langle \vec{S} \rangle(z)|) = \underbrace{(NA dz)}_{\text{total number of scatterers}} \cdot \underbrace{\sigma_{\text{Th}}(|\langle \vec{S} \rangle(z)|)}_{\text{power radiated per scatterer}}$$

We can solve this differential equation by separating variables and integrating:

$$\frac{d(|\langle \vec{S} \rangle(z)|)}{|\langle \vec{S} \rangle(z)|} = -N\sigma_{\text{Th}} dz \implies |\langle \vec{S} \rangle(z)| = |\langle \vec{S} \rangle(0)| e^{-N\sigma_{\text{Th}} z}$$

Since $|\langle \vec{S} \rangle(0)|$ is equal to the intensity of the incident wave $\langle \vec{S} \rangle_{\text{inc}}$, we can rewrite this as:

$$|\langle \vec{S} \rangle(z)| = |\langle \vec{S} \rangle_{\text{inc}}| e^{-N\sigma_{\text{Th}} z} \quad (\text{decay of intensity due to scattering})$$

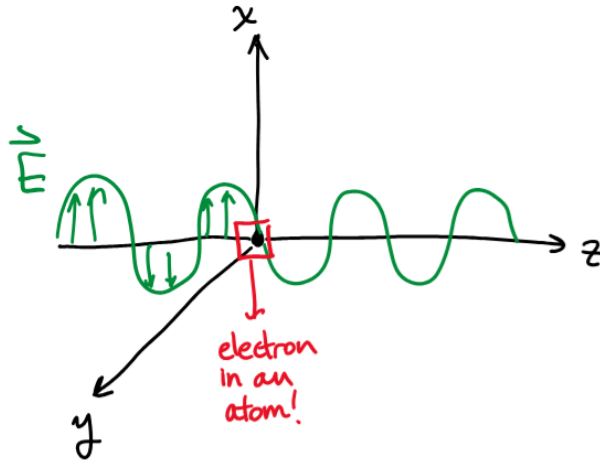
Note: This makes intuitive sense because an increase in the number of free charges, the thickness of the material, or the scattering cross section would all increase the likelihood of scattering, thus leading to a greater decrease in the intensity of the transmitted wave.

Scattering by Bound Charges

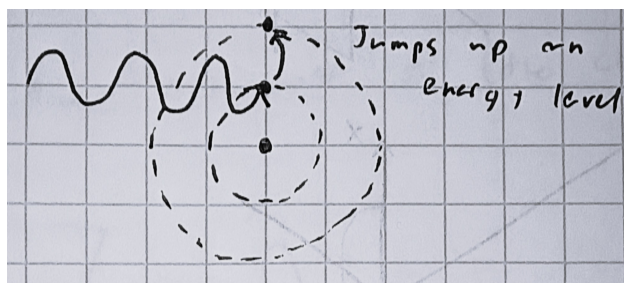
Similar to Thomson scattering, Rayleigh scattering describes the scattering of an electromagnetic wave by a *bound* charge. Instead of a free charge that moves freely in response to the incident wave, a bound charge experiences a restoring force due to its attraction with the nucleus. For our model, we will use Bohr's model of the atom, in which electrons occupy fixed circular orbitals around the nucleus, with radii determined by their energy levels.

Consider an incoming electromagnetic wave with the electric field:

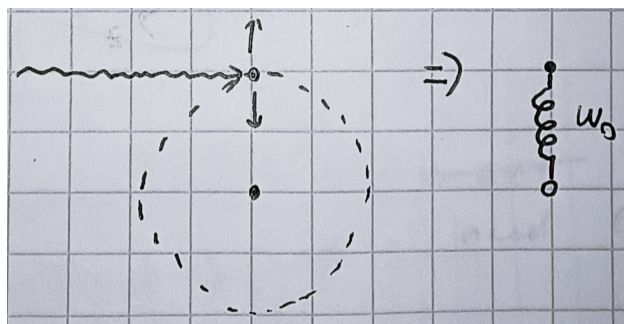
$$\vec{E} = E_0 \cos(\omega t - kz) \hat{x} \implies \vec{E} = E_0 \cos(\omega t) \hat{x} \quad (\text{at origin, } z=0)$$



Recall that in the Bohr model, if a valence electron receives enough energy from the incoming wave, it can be excited to a higher energy level or even ionized (escapes the atom completely).



However, if the incoming wave does not have enough energy to excite the electron to a higher energy level, the electron will simply oscillate in its current orbital in response to the electric field of the wave. For Rayleigh scattering, we are interested in this case, where the valence electron behaves like a driven harmonic oscillator with a natural frequency ω_0 based on the *characteristic frequency* of the atom.



Note: To simplify the model, we are only considering the acceleration of *one* electron in the atom, which follows the valence electron idea.

For a classical simple harmonic oscillator, the characteristic frequency is given as

$$\omega_0 = \sqrt{\frac{k}{m}}$$

We will use this characteristic frequency in our model instead of introducing a stiffness k , since it is more intuitive in the context of atomic behavior. Specifically, the **characteristic** or **transition** frequency ω_0 of an atom is the frequency of the electromagnetic wave that the atom emits or absorbs when it transitions between energy levels. It is often specified in terms of wavelength, but because it corresponds to an electromagnetic wave, we can express it as a frequency using the relation $c = \omega/k$.

Note: If the frequency of the incoming wave ω is close to the characteristic frequency ω_0 , the electron undergoes resonance and may transition to a higher energy level, which is not the behavior we are modeling. As such in the derivation, the condition $\omega = \omega_0$ produces a singularity in the solution.

Solving for the Acceleration

The equation of motion for the driven harmonic oscillator is given by:

$$\sum F_x = m \frac{d^2 x}{dt^2} = \underbrace{qE_0 \cos(\omega t)}_{\vec{E} \text{ field}} - \underbrace{\omega_0^2 m x}_{\text{restoring force}}$$

$$\frac{d^2 x}{dt^2} + \omega_0^2 x = \frac{qE_0}{m} \cos(\omega t)$$

Solving the differential equation for the steady-state particular solution gives the position of the electron as a function of time (guessing a solution of the form $x(t) = A \cos(\omega t)$, or using laplace transform with zero initial conditions):

$$x(t) = \frac{qE_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

$$\implies a(t) = \frac{d^2 x}{dt^2} = \frac{qE_0}{m} \left(\frac{\omega^2}{\omega^2 - \omega_0^2} \right) \cos(\omega t)$$

Larmor Power

Plugging this acceleration into the formula for Larmor power gives the instantaneous power radiated by the accelerating charge:

$$P(t) = \frac{q^2 a^2}{6\pi\epsilon_0 c^3} = \frac{q^4 E_0^2}{6\pi\epsilon_0 m^2 c^3} \left(\frac{\omega^2}{\omega^2 - \omega_0^2} \right)^2 \cos^2(\omega t) \quad (\text{Watts})$$

$$\implies \langle P_{\text{scatt}}(t) \rangle = \underbrace{\frac{8\pi}{3} \left(\frac{q^2}{4\pi\epsilon_0 m c^2} \right)^2}_{\sigma_{\text{Th}}} \left[\frac{\omega^2}{\omega^2 - \omega_0^2} \right]^2 |\langle \vec{S} \rangle_{\text{inc}}|$$

Where the *Rayleigh cross-section area* is simply a rescaling of the Thomson cross-section area by a factor

$$\sigma_{\text{Ray}} = \sigma_{\text{Th}} \left[\frac{\omega^2}{\omega^2 - \omega_0^2} \right]^2$$

$$\boxed{\langle P_{\text{scatt}}(t) \rangle = \sigma_{\text{Ray}} |\langle \vec{S} \rangle_{\text{inc}}|}$$

Limits

In the limit where the incoming has such a high frequency that $\omega \gg \omega_0$, the restoring force becomes negligible, and thus the bound charge behaves like a free charge:

$$\omega \gg \omega_0 \implies \sigma_{\text{Ray}} = \sigma_{\text{Th}}$$

In the limit where the incoming wave has such a low frequency that $\omega \ll \omega_0$, the restoring force dominates, and is thus in the Rayleigh scattering regime:

$$\omega \ll \omega_0 \implies \sigma_{\text{Ray}} = \sigma_{\text{Th}} \left(\frac{\omega}{\omega_0} \right)^4$$

Why is the Sky Blue?

Using Rayleigh scattering, we can explain why the sky is blue. We can do this by comparing the scattered power of blue versus red incident light.

$$\lambda_{\text{blue}} \approx 400 \text{ nm} \quad \lambda_{\text{red}} \approx 700 \text{ nm}$$

Since these are light waves, we can find the corresponding frequencies using

$$\omega = ck = \frac{2\pi c}{\lambda}$$

Notice that frequency is inversely proportional to wavelength, so blue light has a higher frequency than red light. Assuming the same intensity for both blue and red incident light, we can find the ratio of the power scattered by blue light to that of red light:

$$\frac{\langle P \rangle_{\text{scatt, blue}}}{\langle P \rangle_{\text{scatt, red}}} = \frac{\sigma_{\text{Ray, blue}} |\langle \vec{S} \rangle_{\text{inc}}|}{\sigma_{\text{Ray, red}} |\langle \vec{S} \rangle_{\text{inc}}|} = \frac{\cancel{\sigma_{\text{Th}}} \left(\frac{\omega_{\text{blue}}}{\omega_0} \right)^4}{\cancel{\sigma_{\text{Th}}} \left(\frac{\omega_{\text{red}}}{\omega_0} \right)^4} = \left(\frac{\omega_{\text{blue}}}{\omega_{\text{red}}} \right)^4 = \left(\frac{\lambda_{\text{red}}}{\lambda_{\text{blue}}} \right)^4 = \left(\frac{700 \text{ nm}}{400 \text{ nm}} \right)^4$$

$$\implies \langle P \rangle_{\text{scatt, blue}} \approx 10 \langle P \rangle_{\text{scatt, red}}$$

Thus, shorter wavelengths of light are scattered more strongly than longer wavelengths. However, since purple light has an even shorter wavelength than blue light, why isn't the sky purple? In reality, the Sun emits significantly less purple light than blue light, and the human eye is also less sensitive to purple wavelengths. As a result, the sky appears blue rather than purple.

Unit 3A: Static Fields in Matter

Previously, we have only studied the behavior of electromagnetic waves in free space, where there are no charges or currents. Now, we will look at light-matter interactions such as reflection and transmission (how light interacts with *bulk* materials and not free particles). We will start by looking at just *static fields* in **linear dielectric materials**.

In a linear dielectric material:

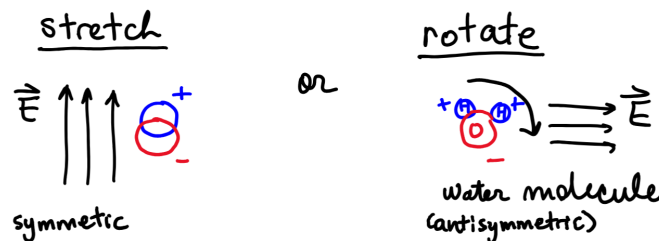
- There is still no free charge or current inside the material, so $\rho = 0$ and $\vec{J} = 0$.
- There will only be effects from bound charges within the material.

Note that because we are looking at static electric and magnetic fields separately, their behavior in matter does NOT represent optical phenomena. In optics, we must consider light as an electromagnetic wave. For now, we are only deriving the fundamental interface conditions for static fields. In Unit 3B, we will introduce the wave nature of light and see how these interface conditions lead to optical phenomena such as reflection and transmission.

Electric Polarization in Materials

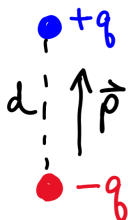
Our goal is to be able to develop a linear model for how a dielectric material responds to an applied electric field. Similar to friction, where the force of friction is proportional to the applied normal force ($F_f = \mu F_n$), we will find that the electric polarization of a material is proportional to the applied electric field.

In a material whose atoms or molecules are bound, there are no mobile free charges (and thus it is not conducting), so under an applied electric field, whether static or oscillating, the atoms or molecules will typically become polarized. This means the positive and negative charges within them are displaced in opposite directions, creating an induced electric dipole moment. This typically occurs through the following mechanisms:



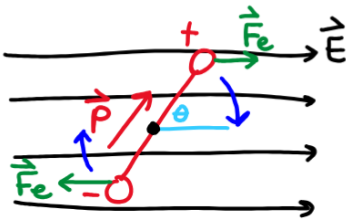
Properties of Electric Dipoles

Recall from Ph-213 that an electric dipole moment $\vec{P}$ is defined as:



$$\vec{p} = q\vec{d} \quad (\text{directed from - to +})$$

Where charges of magnitude q is separated with a displacement d .



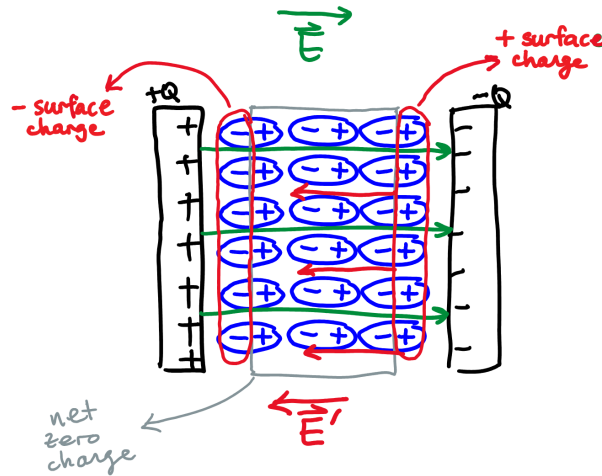
Under an external electric field $\vec{E}$, the dipole experiences a torque $\vec{\tau}$ that tends to align it with the field:

$$\vec{\tau} = \vec{p} \times \vec{E}$$

It is most stable, and thus experiences no torque, when $\vec{p} \parallel \vec{E}$ (angle $\theta = 0$).

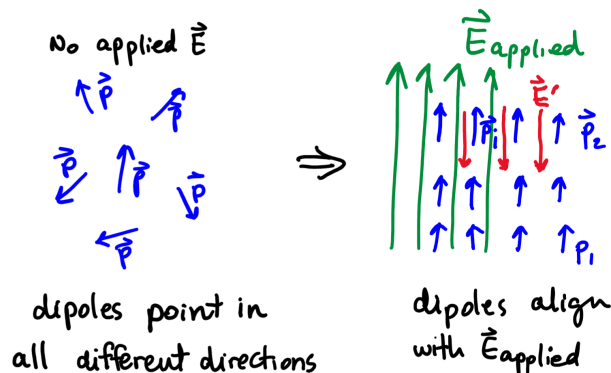
Properties of Dielectrics

Dielectric materials contain atoms or molecules that behave like electric dipoles under an applied electric field. This follows the same intuition as dielectrics in capacitors: the material becomes polarized and induces an *opposing* secondary electric field $\vec{E}'$, which reduces the net electric field strength within the capacitor (and thus increases the capacitance).



This effect is due to the dipoles aligning with the applied electric field, creating surface charges on the dielectric that produce the opposing field $\vec{E}'$. The more *polarizable* the material, the stronger this effect, and therefore the stronger the induced field $\vec{E}'$.

We won't go into the details of the mechanism of polarization. Instead, we want to develop a general simplified model for how the effect of polarization *augments* the applied electric field. Let's begin by quantifying the net effect of an applied electric field on a dielectric material:



Each atom/molecule within the dielectric material has an induced dipole moment $\vec{p}_i$. Thus, we can define a **polarization field** ($\vec{P}$) that describes the *net electric dipole moment per unit volume*:

$$\vec{P} = n \langle \vec{p}_i \rangle$$

Where n is the number density of dipoles (# dipole per m^3) and $\langle \vec{p}_i \rangle$ is the average dipole moment of each atom/molecule.

Note: The average dipole moment $\langle \vec{p}_i \rangle$ depends on the applied electric field. When the dipole moments are more random, the average dipole moment is smaller, whereas when the dipole moments are more aligned, the average dipole moment is larger.

Relating Polarization to Electric Field

Using unit analysis, we can model $\vec{P}$ as proportional to $\vec{E}_{\text{applied}}$:

$$\vec{P} = [?] \vec{E}$$

Unit analysis:

$$\begin{aligned} \vec{P} &= \left[\frac{1}{\text{m}^3} \cdot \text{C} \cdot \text{m} \right] = \left[\frac{\text{C}}{\text{m}^2} \right] \\ \vec{E} &= \left[\frac{\text{N}}{\text{C}} \right] \\ \implies [?] &= \frac{\text{C}^2}{\text{N} \cdot \text{m}^2} \sim \epsilon_0 \text{ (permittivity)} \end{aligned}$$

Therefore, we can model the polarization field as being proportional to the applied electric field with a proportionality constant of ϵ_0 :

$$\boxed{\vec{P} = \epsilon_0 \chi_e \vec{E}}$$

Where χ_e is the **dielectric susceptibility** (unitless, pronounced "chi"). It quantifies how much a material can become polarized in an applied $\vec{E}$ field (polarizability of the material).

For a *linear dielectric material*, χ_e is a constant, and thus the polarization field $\vec{P}$ is linearly proportional to the applied electric field $\vec{E}$.

Maxwell's Equations in Dielectric Materials

Applying Gauss's law to a dielectric material, we find that we have to consider both the free charge and the bound charge:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_{\text{total}}}{\epsilon_0} \implies \vec{\nabla} \cdot \underbrace{(\epsilon_0 \vec{E})}_{\substack{\text{same units} \\ \text{as } \vec{P}}} = \rho_{\text{free}} + \rho_{\text{bound}}$$

This is saying that the behavior (divergence) of the electric field is determined by both the free charge and the bound charge. However, we have developed the polarization field $\vec{P}$

precisely to describe the effect of the bound charge on the electric field. Therefore, we can rewrite the equation as:

$$\vec{\nabla} \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho_{\text{free}}$$

Note: The reason we can do this is because $\vec{\nabla} \cdot \vec{P} = -\rho_{\text{bound}}$, which basically says that whenever the polarization field is non-uniform, it implies that dipoles are not perfectly canceling, leading to an effective charge density. Regions where $\vec{P}$ diverges act like sources of negative bound charge, while regions where it converges correspond to positive bound charge.

To make life easier, we define the **electric displacement field** $\vec{D}$ as:

$$\begin{aligned} \vec{D} &= \epsilon_0 \vec{E} + \vec{P} \\ &= \epsilon_0 \vec{E} + \epsilon_0 \chi_e \vec{E} = \epsilon_0 \underbrace{(1 + \chi_e)}_{K: \text{dielectric constant}} \vec{E} = \underbrace{\epsilon_0 K}_{\epsilon} \vec{E} = \epsilon \vec{E} \end{aligned}$$

$$\boxed{\vec{D} = \epsilon \vec{E}}$$

Where ϵ is the **permittivity** of the material, which quantifies how strongly the material polarizes in response to an electric field (and thus how much it reduces the net electric field).

$$\implies \vec{\nabla} \cdot \vec{D} = \rho_{\text{free}} \quad \text{or} \quad \vec{\nabla} \cdot (\epsilon \vec{E}) = \rho_{\text{free}} \quad \text{or} \quad K \int \vec{E} \cdot d\vec{A} = \frac{q_{\text{free}}}{\epsilon_0}$$

In a dielectric, there is no free charge, so $\rho_{\text{free}} = 0$:

$$\boxed{\vec{\nabla} \cdot \vec{D} = 0 \quad (\text{Gauss's law in a linear dielectric})}$$

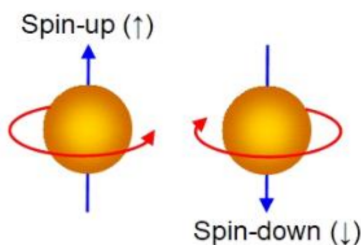
Understanding the Displacement Field

The electric field $\vec{E}$ is the *fundamental field*, it accounts for the total effect of both free and bound charges. Thus, because bound charges are present in a material, the $\vec{E}$ field changes as it passes through a material.

The displacement field $\vec{D}$ is a *derived field*, it accounts for ONLY the effect of free charges (as seen in Gauss's law in a linear dielectric). Thus, because there are no free charges in a material, the $\vec{D}$ field does not change as it passes through a material (it can still change at the boundary).

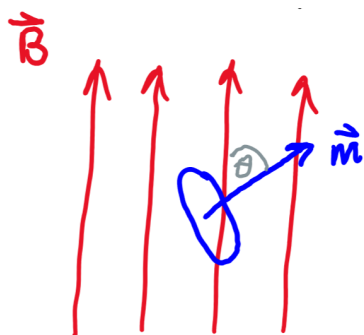
Magnetization in Materials

Analogous to electric polarization, fundamental units of matter have an *intrinsic magnetic dipole moment* called **spin**, which can align with an applied magnetic field. This magnetic dipole moment is similar to the dipole moment of a current loop and bar magnet, thus we can apply the same intuition of before to quantify the net effect of an applied magnetic field on a material.



Properties of Magnetic Dipoles

In Ph-213, we defined the magnetic dipole moment $\vec{m}$ for current loops. For spins, we will not derive it here and simply denote their magnetic dipole moment as $\vec{m}$.



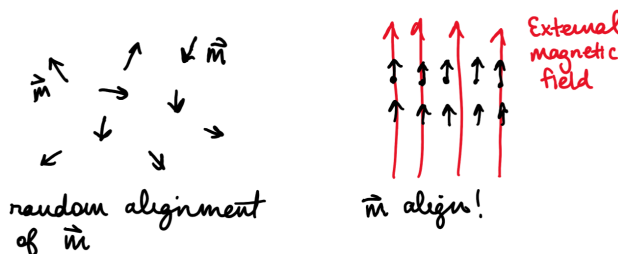
Under an external magnetic field $\vec{B}$, the dipole experiences a torque $\vec{\tau}$ that tends to align it with the field:

$$\vec{\tau} = \vec{m} \times \vec{B}$$

It is most stable, and thus experiences no torque, when $\vec{m} \parallel \vec{B}$ (angle $\theta = 0$).

Properties of Linear Magnetic Materials

In exactly the same logic as how we derived the polarization field $\vec{P}$ for electric dipoles, we can derive a **magnetization field** ($\vec{M}$) for magnetic dipoles.



$$\vec{M} = n \langle \vec{m}_i \rangle$$

Where n is the number density of magnetic dipoles (# dipole per m^3) and $\langle \vec{m}_i \rangle$ is the average magnetic dipole moment of each atom/molecule.

Since this is a linear magnetic material, we can model the magnetization field as being proportional to the applied magnetic field with using the magnetic susceptibility χ_m :

$$\boxed{\vec{M} = \chi_m \vec{H}}$$

Here, we introduce the magnetic field intensity $\vec{H}$, which can be thought of a more "fundamental" field than the magnetic field $\vec{B}$ purely mathematically. Historically, magnetism was studied using permanent magnets, meaning it was initially understood in terms of magnetic fields in matter.

Maxwell's Equations in Magnetic Materials

Applying Ampere-Maxwell's law to a magnetic material, we find that we have to consider both the free current and both bound currents (from magnetization and time-varying polarization) $(\frac{\partial \vec{P}}{\partial t})$, and the displacement current associated with the changing electric field $(\epsilon_0 \frac{\partial \vec{E}}{\partial t})$.

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}_{\text{total}} + \mu_0 \frac{\partial}{\partial t}(\epsilon_0 \vec{E} + \vec{P}) \implies \vec{\nabla} \times \underbrace{\left(\frac{1}{\mu_0} \vec{B} \right)}_{\substack{\text{same units} \\ \text{as } \vec{M}}} = \vec{J}_{\text{free}} + \vec{J}_{\text{bound}} + \frac{\partial}{\partial t}(\epsilon_0 \vec{E} + \vec{P})$$

Like how we used the polarization field $\vec{P}$ to account for the effect of bound charges on the electric field, we can use the magnetization field $\vec{M}$ to account for the effect of bound currents on the magnetic field. Therefore, we can rewrite the equation as:

$$\vec{\nabla} \times \left(\frac{1}{\mu_0} \vec{B} - \vec{M} \right) = \vec{J}_{\text{free}} + \frac{\partial}{\partial t}(\epsilon_0 \vec{E} + \vec{P})$$

Like before when we defined the electric displacement field $\vec{D}$, we can define the **magnetic field intensity** $\vec{H}$ to simplify the equation:

$$\begin{aligned} \vec{H} &= \frac{1}{\mu_0} \vec{B} - \vec{M} \\ \implies \vec{B} &= \mu_0 \vec{H} + \mu_0 \vec{M} = \mu_0 \underbrace{(1 + \chi_m)}_{\mu_r} \vec{H} \\ &= \mu_0 \mu_r \vec{H} \end{aligned}$$

$$\boxed{\vec{B} = \mu \vec{H}}$$

Where μ is the **permeability** of the material, which quantifies how strongly the material magnetizes in response to a magnetic field and μ_r is the **relative permeability** of the material.

$$\implies \vec{\nabla} \times \vec{H} = \vec{J}_{\text{free}} + \frac{\partial}{\partial t}(\epsilon_0 \vec{E} + \vec{P})$$

Since there are no free currents inside a magnetic material, $\vec{J}_{\text{free}} = 0$, and rewriting in terms of $\vec{D}$ gives:

$$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \quad (\text{Ampere-Maxwell's law in a linear magnetic material})$$

Note: This section is peak mathematical gymnastics where we are using all our simplifying fields ($\vec{P}$, $\vec{D}$, $\vec{M}$, $\vec{H}$) to clean up our equations. Honestly, I would have liked to follow the same logic as the electric polarization section and defined $\vec{M}$ in terms of $\vec{B}$ and only $\vec{H}$ after we derived the Maxwell's equations in magnetic materials, but it just makes the math messier. The key is that the *physics* never changed, we just introduced these linear models and simplifying fields to make our equations more intuitive and easier to work with.

Units:

$$\vec{H} : \left[\frac{\text{A}}{\text{m}} \right]$$

$$\vec{B} : [\text{T}] \quad \text{or} \quad \left[\frac{\text{Wb}}{\text{m}^2} \right] \quad \text{or} \quad \left[\frac{\text{V} \cdot \text{s}}{\text{m}^2} \right]$$

Understanding the Magnetic Field Intensity

Looking at Ampere-Maxwell's law in a linear magnetic material alone gets messy since we are considering the effects of free currents, bound currents, and induced currents from changing electric fields. However, we can still gain some intuition about the $\vec{B}$ and $\vec{H}$ fields. While we call $\vec{H}$ the fundamental field, $\vec{B}$ is actually the field that accounts for ALL the physical effects of the currents and changing electric fields. As such, it also accounts for the effect of magnetization of the material, and changes as it passes through a material (like $\vec{E}$).

On the other hand, $\vec{H}$ behaves more like a derived field that only accounts for the effect of free currents, and thus does not change as it passes through a material (like $\vec{D}$).

Just to say, we're free to think of $\vec{H}$ and $\vec{D}$ however we want, as long as we understand the reason WHY we defined them in the first place. Thus, we can think that $\vec{H}$ is *mathematically* analogous to $\vec{E}$ and $\vec{B}$ is *mathematically* analogous to $\vec{D}$ (while $\vec{B}$ is the physical field and $\vec{H}$ is a simplifying field like $\vec{D}$, so the analogy is not perfect). For now, I'll continue with the analogy of $\vec{H}$ and $\vec{E}$ being more fundamental purely because it makes the math cleaner.

Maxwell's Equations in Matter

Using the equations we derived for $\vec{D}$ and $\vec{H}$, we can now write Maxwell's equations in linear dielectric and magnetic materials. While we assume there are no free charges or currents in the material, I will still include them in the equations for completeness.

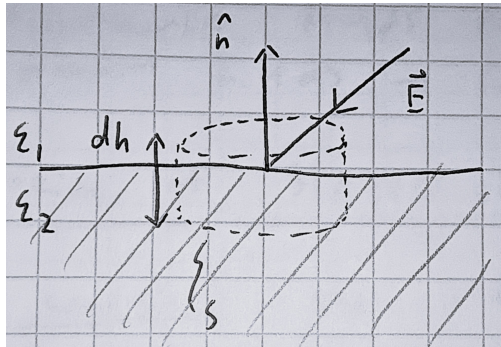
$\begin{aligned}\vec{\nabla} \cdot \vec{D} &= \rho_{\text{free}} & \vec{\nabla} \cdot \vec{B} &= 0 \\ \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} & \vec{\nabla} \times \vec{H} &= \vec{J}_{\text{free}} + \frac{\partial \vec{D}}{\partial t}\end{aligned}$	Where: $\begin{aligned}\vec{D} &= \epsilon \vec{E} \\ \vec{B} &= \mu \vec{H}\end{aligned}$
--	---

Assumption: $\rho_{\text{free}} = 0$ and $\vec{J}_{\text{free}} = 0$ inside the material.

Interface Conditions

Having developed the $\vec{D}$ and $\vec{H}$ fields to model the bulk response of materials to electromagnetic fields, we can now derive the interface conditions for static fields without having to account for bound charges and currents by applying Maxwell's equations at the boundary.

Note: Although real surfaces are bumpy and irregular, at a *small* scale, they can be approximated as flat planes. **Applying Gauss's Law for Electricity at the Interface**
To describe the behavior of the electric and displacement fields at the boundary, we define a Gaussian pillbox and take the limit that the thickness of the pillbox goes to zero so that it zooms in on the interface.



$$\vec{\nabla} \cdot \vec{D} = 0 \rightarrow \oint_S \vec{D} \cdot d\vec{a} = 0 \quad (\text{Divergence Thm})$$

$$\lim_{dh \rightarrow 0} \left[\oint_S \vec{D} \cdot d\vec{a} \right] = 0$$

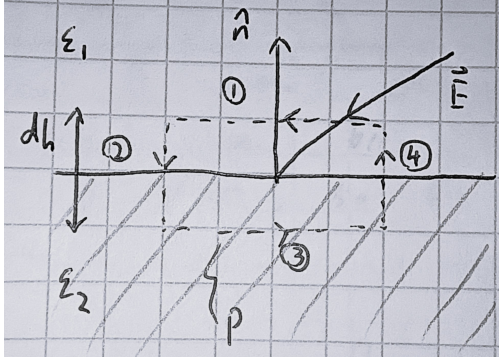
$$\Rightarrow \int_{\text{Top}} \vec{D} \cdot d\vec{a} + \int_{\text{Bot}} \vec{D} \cdot d\vec{a} + \underbrace{\int_{\text{Side}} \vec{D} \cdot d\vec{a}}_{0 \text{ as } dh \rightarrow 0} = 0$$

$$\Rightarrow \underbrace{-D_1^\perp}_{\text{entering}} + \underbrace{D_2^\perp}_{\text{exiting}} = 0$$

Where $D_1^\perp$ and $D_2^\perp$ are the components of $\vec{D}$ along $\hat{n}$ (normal to the interface) in medium 1 and medium 2, respectively.

$$\Rightarrow D_1^\perp = D_2^\perp, \quad \epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp$$

Applying Faraday's Law at the Interface



$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \rightarrow \oint_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} = \oint_S \left(-\frac{\partial \vec{B}}{\partial t} \right) \cdot d\vec{a}$$

$$\rightarrow \oint_P \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{a}$$

(Stokes thm) (constant fixed surface)

$$\lim_{dh \rightarrow 0} \left[\oint_P \vec{E} \cdot d\vec{l} \right] = \lim_{dh \rightarrow 0} \left[-\frac{\partial}{\partial t} \underbrace{\int_S \vec{B} \cdot d\vec{a}}_{\substack{\text{flux goes to} \\ \text{zero as } dh \rightarrow 0}} \right]$$

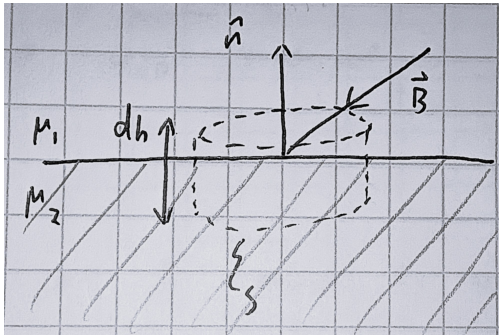
$$\Rightarrow \int_1 \vec{E} \cdot d\vec{l} + \underbrace{\int_2 \vec{E} \cdot d\vec{l}}_0 + \int_3 \vec{E} \cdot d\vec{l} + \underbrace{\int_4 \vec{E} \cdot d\vec{l}}_0 = 0 \Rightarrow \underbrace{E_1^{\parallel}}_{\text{parallel}} - \underbrace{E_2^{\parallel}}_{\text{antiparallel}} = 0$$

Where $E_1^{\parallel}$ and $E_2^{\parallel}$ are the components of $\vec{E}$ parallel to the interface in medium 1 and medium 2, respectively.

$$\Rightarrow E_1^{\parallel} = E_2^{\parallel}, \quad \frac{D_1^{\parallel}}{\varepsilon_1} = \frac{D_2^{\parallel}}{\varepsilon_2}$$

Now we apply the same intuition to derive the interface conditions for the magnetic field.

Applying Gauss's Law for Magnetism at the Interface



$$\vec{\nabla} \cdot \vec{B} = 0 \rightarrow \oint_S \vec{B} \cdot d\vec{a} = 0 \quad (\text{Divergence Thm})$$

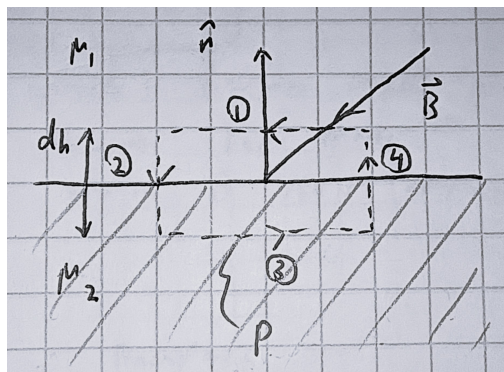
$$\lim_{dh \rightarrow 0} \left[\oint_S \vec{B} \cdot d\vec{a} \right] = 0$$

$$\Rightarrow \int_{\text{Top}} \vec{B} \cdot d\vec{a} + \int_{\text{Bot}} \vec{B} \cdot d\vec{a} + \underbrace{\int_{\text{Side}} \vec{B} \cdot d\vec{a}}_{0 \text{ as } dh \rightarrow 0} = 0$$

$$\Rightarrow \underbrace{-B_1^{\perp}}_{\text{entering}} + \underbrace{B_2^{\perp}}_{\text{exiting}} = 0$$

$$\Rightarrow B_1^{\perp} = B_2^{\perp}, \quad \mu_1 H_1^{\perp} = \mu_2 H_2^{\perp}$$

Applying Ampere-Maxwell Law at the Interface



$$\begin{aligned}\vec{\nabla} \times \vec{H} &= \frac{\partial \vec{D}}{\partial t} \rightarrow \oint_S (\vec{\nabla} \times \vec{H}) \cdot d\vec{a} = \oint_S \left(\frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{a} \\ &\rightarrow \oint_P \vec{H} \cdot d\vec{l} = \frac{\partial}{\partial t} \oint_S \vec{D} \cdot d\vec{a} \\ &\quad \text{(Stokes thm)} \quad \quad \text{(constant fixed surface)}\end{aligned}$$

$$\begin{aligned}\lim_{dh \rightarrow 0} \left[\oint_P \vec{H} \cdot d\vec{l} \right] &= \lim_{dh \rightarrow 0} \left[\frac{\partial}{\partial t} \underbrace{\oint_S \vec{D} \cdot d\vec{a}}_{\text{flux goes to zero as } dh \rightarrow 0} \right] \\ \Rightarrow \int_1 \vec{H} \cdot d\vec{l} + \underbrace{\int_2 \vec{H} \cdot d\vec{l}}_0 + \int_3 \vec{H} \cdot d\vec{l} + \underbrace{\int_4 \vec{H} \cdot d\vec{l}}_0 &= 0 \Rightarrow \underbrace{H_1^{\parallel}}_{\text{parallel}} - \underbrace{H_2^{\parallel}}_{\text{antiparallel}} = 0\end{aligned}$$

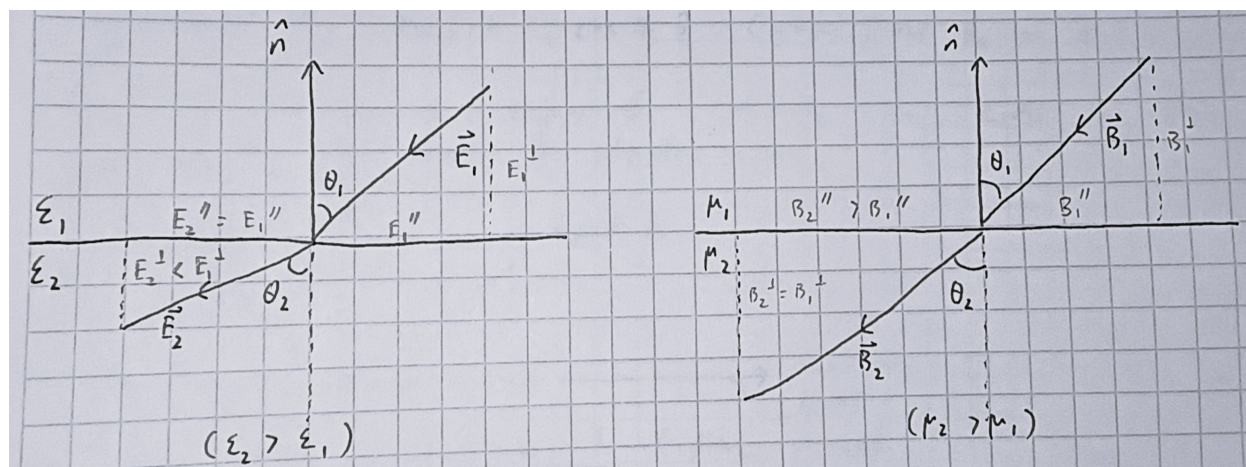
$$\boxed{\Rightarrow H_1^{\parallel} = H_2^{\parallel}, \quad \frac{B_1^{\parallel}}{\mu_1} = \frac{B_2^{\parallel}}{\mu_2}}$$

Summary of Interface Conditions

We can rewrite the components of the fields in terms of $\hat{n}$ by using dot products for the perpendicular components and cross products for the parallel components:

$\left. \begin{aligned} (1) \quad \hat{n} \cdot (\vec{D}_1 - \vec{D}_2) &= 0 \\ (2) \quad \hat{n} \cdot (\vec{B}_1 - \vec{B}_2) &= 0 \end{aligned} \right\}$	Normal components of $\vec{D}$ & $\vec{B}$ are continuous	$D_1^\perp = D_2^\perp \implies \epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp$ $B_1^\perp = B_2^\perp \implies \mu_1 H_1^\perp = \mu_2 H_2^\perp$
$\left. \begin{aligned} (3) \quad \hat{n} \times (\vec{E}_1 - \vec{E}_2) &= 0 \\ (4) \quad \hat{n} \times (\vec{H}_1 - \vec{H}_2) &= 0 \end{aligned} \right\}$	Tangential components of $\vec{E}$ & $\vec{H}$ are continuous	$E_1^\parallel = E_2^\parallel \implies \frac{D_1^\parallel}{\epsilon_1} = \frac{D_2^\parallel}{\epsilon_2}$ $H_1^\parallel = H_2^\parallel \implies \frac{B_1^\parallel}{\mu_1} = \frac{B_2^\parallel}{\mu_2}$

Visually, it looks like the bending of the electric and magnetic fields at the interface:



Note that the magnitudes of the fields can also change at the interface, not just the directions. For example, looking at the magnitudes of the electric fields:

$$|\vec{E}_1| = \sqrt{(E_1^\parallel)^2 + (E_1^\perp)^2}$$

$$|\vec{E}_2| = \sqrt{(E_2^\parallel)^2 + (E_2^\perp)^2}$$

Since $E_2^\perp \neq E_1^\perp$, the magnitudes of the electric fields must be different! When applied to light in unit 3B, this will lead to the conclusion that the speed of light changes in different materials.

Unit 3B: Optics

In this unit, we will be applying the interface conditions we derived previously to electromagnetic waves, and thus derive the laws of optics such as reflection and refraction. We will start by looking at the properties of $\vec{E}$, $\vec{B}$, $\vec{D}$, and $\vec{H}$ fields as oscillating waves instead of static fields.

Electromagnetic Waves in Matter

Since $\vec{D}$ and $\vec{H}$ are just rescaled versions of $\vec{E}$ and $\vec{B}$, we can model them using the same plane wave approximation as before. Recall that the general plane wave solution is in the form:

$$\vec{E}(\vec{k}, t) = (E_{0x}e^{i\phi_x}\hat{x} + E_{0y}e^{i\phi_y}\hat{y} + E_{0z}e^{i\phi_z}\hat{z}) e^{i(\vec{k}\cdot\vec{r}-\omega t)} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)}$$

Note that all the fields share the same wave properties ($\vec{k}$, ω , and thus v), since they make up the same electromagnetic wave.

Additionally, we can show using Maxwell's equations that all the fields satisfy the 3D wave equation:

$$\nabla^2 \vec{E} = \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2} \quad \nabla^2 \vec{H} = \mu\epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

$$\nabla^2 \vec{D} = \mu\epsilon \frac{\partial^2 \vec{D}}{\partial t^2} \quad \nabla^2 \vec{B} = \mu\epsilon \frac{\partial^2 \vec{B}}{\partial t^2}$$

Note that the speed of the wave in a material is ALWAYS less than the speed of light in vacuum, since $\mu\epsilon > \mu_0\epsilon_0$ (since $K > 1$ and $\mu_r > 1$ for all materials):

$$v = \frac{1}{\sqrt{\mu\epsilon}} < \frac{1}{\sqrt{\mu_0\epsilon_0}} = c$$

Additional Properties

- $\vec{E}$, $\vec{B}$, $\vec{D}$, and $\vec{H}$ are all perpendicular to the direction of wave propagation $\hat{k}$ (transverse waves).
- $\vec{E}$ and $\vec{D}$ are perpendicular to $\vec{B}$ and $\vec{H}$, and thus are parallel to each other.
- $\frac{|\vec{E}|}{|\vec{H}|} = \sqrt{\frac{\mu}{\epsilon}}$ (wave impedance)

We define **wave impedance** (Z) as a material property that tells us about the transmission of electromagnetic waves through a material (ex: impedance matching).

$$Z = \frac{|\vec{E}|}{|\vec{H}|} = \sqrt{\frac{\mu}{\epsilon}}$$

Note: This is derived from applying Faraday's law.

Poynting Vector in Matter

We can rewrite Poynting vector in terms of our "fundamental" fields $\vec{E}$ and $\vec{H}$:

$$u_E = \frac{1}{2}\epsilon \vec{E} \cdot \vec{E}, \quad u_B = \frac{1}{2}\mu \vec{H} \cdot \vec{H}$$

Thus the total energy density of the electromagnetic wave is:

$$u = u_E + u_B = \frac{1}{2}(\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H})$$

$$\boxed{\vec{S} = \vec{E} \times \vec{H} \quad \implies \quad |\vec{S}| = |\vec{E}||\vec{H}| = \frac{|\vec{E}|^2}{Z}}$$

The lower the wave impedance of a material, the higher the energy flux of the electromagnetic wave through the material for a given electric field amplitude.

Note that Poynting vector is defined for any electromagnetic field, not just for light and oscillating fields.

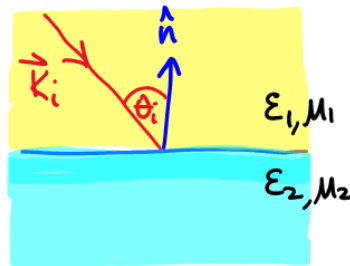
Recall: The Poynting vector is derived from applying Maxwell's equations to get the form of the continuity equation:

$$\frac{\partial u}{\partial t} + \vec{\nabla} \cdot \vec{S} = 0$$

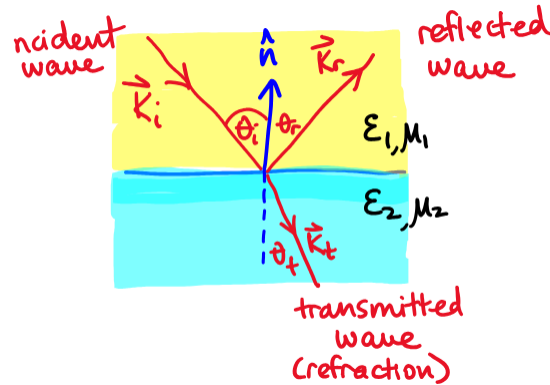
However, deriving the energy densities directly is a bit more involved since we cannot just use the same energy density equations for electric and magnetic fields in vacuum.

Optics Set-Up and Definitions

Plane of Incidence: the formed by the incident wavevector $\vec{k}_i$ and the normal vector $\hat{n}$ to the interface. $\vec{k}_i$ can be in any general direction, but typically we will take it to be within the $x - y$ or $y - z$ plane to simplify the math.



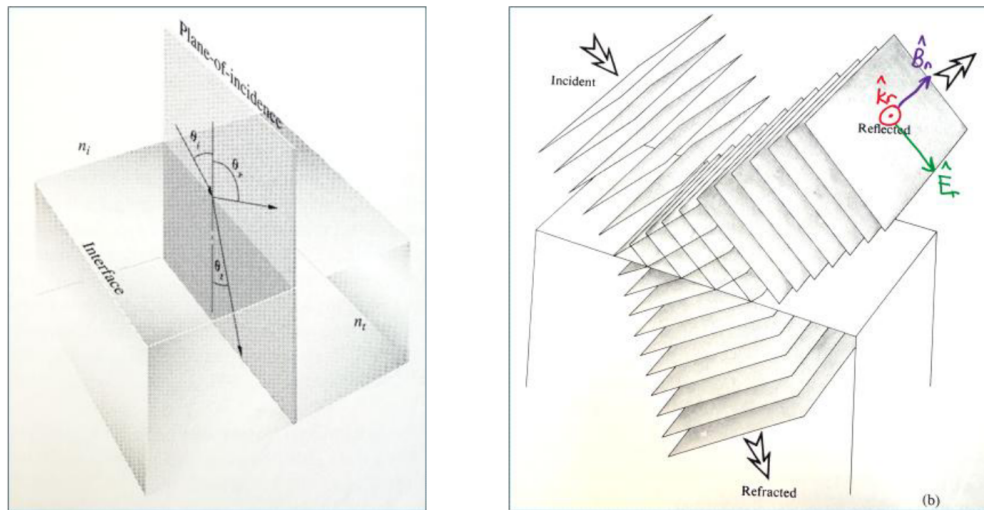
Once a wave is incident at an interface, there can be a **reflected wave**, a **transmitted wave**, or both.



Assumptions:

- Both the reflected and transmitted waves exist within the same plane of incidence as the incident wave.
- Upon reflection, the electric field undergoes a phase shift of 180° . This typically occurs when light reflects off a medium with a higher refractive index. If we are consistent with this assumption, we will sometimes find a negative sign for the reflected electric field, which indicates that we assumed the wrong direction (just like getting a negative force in a FBD).

In 3D, this looks like:



Each EM wave (incident, reflected, transmitted) will have an oscillating $\vec{E}$ and $\vec{B}$ field, each having a plane wave solution:

$$\begin{aligned}
 \text{incident: } \vec{E}_i &= \vec{E}_{0i} e^{i(\vec{k}_i \cdot \vec{r} - \omega_i t)}, & \vec{B}_i &= \vec{B}_{0i} e^{i(\vec{k}_i \cdot \vec{r} - \omega_i t)} \\
 \text{reflected: } \vec{E}_r &= \vec{E}_{0r} e^{i(\vec{k}_r \cdot \vec{r} - \omega_r t)}, & \vec{B}_r &= \vec{B}_{0r} e^{i(\vec{k}_r \cdot \vec{r} - \omega_r t)} \\
 \text{transmitted: } \vec{E}_t &= \vec{E}_{0t} e^{i(\vec{k}_t \cdot \vec{r} - \omega_t t)}, & \vec{B}_t &= \vec{B}_{0t} e^{i(\vec{k}_t \cdot \vec{r} - \omega_t t)}
 \end{aligned}$$

We still start off by generalizing the amplitudes, wavevectors, and frequencies of each wave, and then we will apply the interface conditions to find the relationships between them.

For our setup, we take the plane of incidence to be the y - z plane and the interface to lie at $z = 0$. Since the waves are plane waves and uniform in the x direction, the position vector reduces to $\vec{r} = y \hat{y} + z \hat{z}$. Thus, the plane wave solutions can be rewritten in as:

$$\vec{E}_i(\vec{r}, t) = \vec{E}_{0i} e^{i(k_i \sin \theta_i y - k_i \cos \theta_i z - \omega_i t)}$$

$$\vec{E}_r(\vec{r}, t) = \vec{E}_{0r} e^{i(k_r \sin \theta_r y + k_r \cos \theta_r z - \omega_r t)}$$

$$\vec{E}_t(\vec{r}, t) = \vec{E}_{0t} e^{i(k_t \sin \theta_t y - k_t \cos \theta_t z - \omega_t t)}$$

We will be ONLY considering the fields at the interface ($z = 0$):

$$\vec{E}_i = \vec{E}_{0i} e^{i(y k_i \sin \theta_i - \omega_i t)}, \quad \vec{E}_r = \vec{E}_{0r} e^{i(y k_r \sin \theta_r - \omega_r t)}, \quad \vec{E}_t = \vec{E}_{0t} e^{i(y k_t \sin \theta_t - \omega_t t)}$$

Note: Instead of writing components of the wavevector as k_y and k_z , we can write them in terms of the angle formed with the normal vector $\hat{n}$ to the interface.

Additionally, note that speed of the wave is dependent on the medium, and thus we have the following relationships between the $\vec{E}$ and $\vec{B}$ fields in each medium:

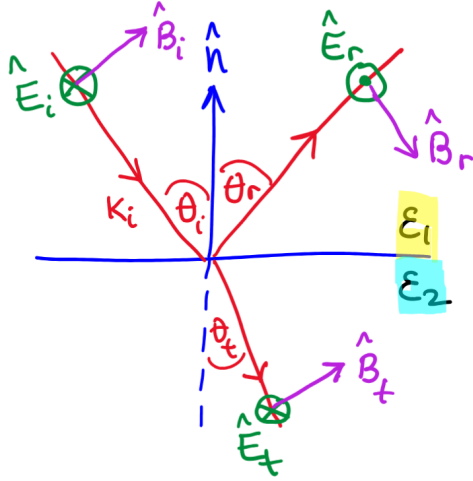
$$\frac{|\vec{E}_{0i}|}{|\vec{B}_{0i}|} = \frac{\omega_i}{k_i} = v_1 = \frac{1}{\sqrt{\mu_1 \epsilon_1}}, \quad \frac{|\vec{E}_{0r}|}{|\vec{B}_{0r}|} = \frac{\omega_r}{k_r} = v_1 = \frac{1}{\sqrt{\mu_1 \epsilon_1}} \quad (\text{in medium 1})$$

$$\frac{|\vec{E}_{0t}|}{|\vec{B}_{0t}|} = \frac{\omega_t}{k_t} = v_2 = \frac{1}{\sqrt{\mu_2 \epsilon_2}} \quad (\text{in medium 2})$$

Two Types of Polarization

The electric field of the wave can be polarized in different directions, and any other general polarization can be decomposed into these two types of polarization.

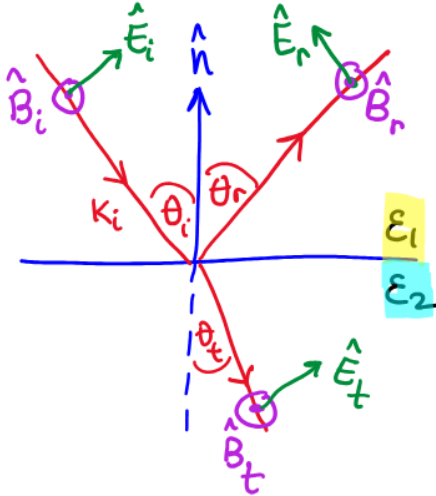
”Case A”: S-Polarization (Perpendicular Polarization)



- $\vec{E}$ fields are $\perp$ to the plane of incidence and thus $\parallel$ to the interface.
- $\vec{B}$ fields are $\parallel$ to the plane of incidence and thus have both $\parallel$ and $\perp$ components to the interface.
- $\vec{E}$, $\vec{B}$, and $\hat{k}$ are all mutually perpendicular to each other.

Note that the phase shift of 180° in the electric field is already encoded in the chosen direction of the reflected field. Like a FBD, the positive directions of each field are drawn, and if we find a negative sign for the respective field then it just means we assumed the wrong direction.

”Case B”: P-Polarization (Parallel Polarization)



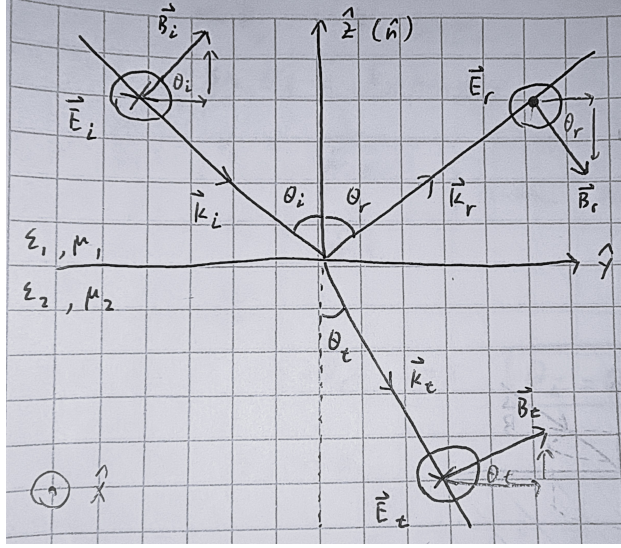
- $\vec{E}$ fields are in the plane of incidence and thus have both $\parallel$ and $\perp$ components to the interface.
- $\vec{B}$ fields are $\perp$ to the plane of incidence and thus $\parallel$ to the interface.
- $\vec{E}$, $\vec{B}$, and $\hat{k}$ are all mutually perpendicular to each other.

Note that the $\vec{E}$ fields are not shifted by 180° upon reflection, this is due to physical intuition of the directions of the fields. As always, if we assumed the wrong direction, then the math will tell us.

Next, our goal is to predict how light will reflect and/or transmit based on our interface conditions and material properties. We will do this by deriving the reflection and transmission coefficients for both cases of polarization, which are called the **Fresnel Formulae**:

$$\begin{aligned}
 E_{0r}^a &= r_{\perp} E_{0i}^a & r_{\perp} &: \text{reflection coefficient (S-Polarization)} \\
 E_{0t}^a &= t_{\perp} E_{0i}^a & t_{\perp} &: \text{transmission coefficient (S-Polarization)} \\
 E_{0r}^b &= r_{\parallel} E_{0i}^b & r_{\parallel} &: \text{reflection coefficient (P-Polarization)} \\
 E_{0t}^b &= t_{\parallel} E_{0i}^b & t_{\parallel} &: \text{transmission coefficient (P-Polarization)}
 \end{aligned}$$

Case A: Fresnel Coefficients (S-Polarization)



Apply interface condition (3) that parallel components of $\vec{E}$ to the interface are continuous:

$$\hat{n} \times (\vec{E}_1 - \vec{E}_2) = 0 \iff E_1^{\parallel} = E_2^{\parallel}$$

This applies to the *total* $\vec{E}$ field in each medium:

$$\underbrace{E_i^{\parallel} + E_r^{\parallel}}_{\text{medium 1}} = \underbrace{E_t^{\parallel}}_{\text{medium 2}}$$

Writing the interface condition in terms of the plane wave solutions at the interface (taking the positive direction to be along $\hat{x}$):

$$\implies (-E_{0i}e^{i(yk_i \sin \theta_i - \omega_i t)}) + (E_{0r}e^{i(yk_r \sin \theta_r - \omega_r t)}) = (-E_{0t}e^{i(yk_t \sin \theta_t - \omega_t t)})$$

Physically, the coefficients $r_{\perp}$ and $t_{\perp}$ must be independent of position and time, since they depend on just material properties. In other words, the interface condition must hold true for *all* y and t . Thus, assuming that the exponentials are equal, we get the result:

$$\boxed{-E_{0i} + E_{0r} = -E_{0t} \quad (\text{IC 3})}$$

We can derive several useful results from our assumption:

- The frequency of the reflected and transmitted light is the same as the frequency of the incident light:

$$\boxed{\omega_i = \omega_r = \omega_t}$$

- We can derive **law of reflection** as well:

$$k_i \sin \theta_i = k_r \sin \theta_r = k_t \sin \theta_t$$

Since $v = \frac{\omega}{k}$, and ω is the same for all light:

$$v_i = \frac{\omega}{k_i}, \quad v_r = \frac{\omega}{k_r}, \quad v_t = \frac{\omega}{k_t}$$

Since the incident and reflected light are in the same medium, they have the same speed $v_1 = \frac{1}{\sqrt{\mu_1 \epsilon_1}}$:

$$v_1 = v_i = v_r \implies k_i = k_r$$

Therefore, we get the law of reflection:

$$k_i \sin \theta_i = k_r \sin \theta_r \implies \sin \theta_i = \sin \theta_r$$

$$\boxed{\theta_i = \theta_r}$$

- Finally, we can derive **Snell's Law (law of refraction)**:

$$k_i \sin \theta_i = k_t \sin \theta_t$$

Since the speed of light is different in the two media, we have:

$$v_1 = \frac{\omega}{k_i} = \frac{1}{\sqrt{\mu_1 \epsilon_1}}, \quad v_2 = \frac{\omega}{k_t} = \frac{1}{\sqrt{\mu_2 \epsilon_2}}$$

Defining the **index of refraction** (n) as:

$$n = \frac{c}{v}$$

Where c is the speed of light in vacuum, and v is the speed of light in the material.

Rewriting the wavevectors in terms of the index of refraction:

$$k_i = \frac{\omega}{v_1} = \frac{\omega}{c} n_1 \quad \text{and} \quad k_t = \frac{\omega}{v_2} = \frac{\omega}{c} n_2$$

Now, $k_i \sin \theta_i = k_t \sin \theta_t$ implies Snell's law:

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

Now continuing by applying interface condition (4) that parallel components of $\vec{H}$ to the interface are continuous:

$$\hat{n} \times (\vec{H}_1 - \vec{H}_2) = 0 \iff H_1^{\parallel} = H_2^{\parallel}$$

Like with the electric field, this applies to the *total* $\vec{H}$ field in each medium:

$$\underbrace{H_i^{\parallel} + H_r^{\parallel}}_{\text{medium 1}} = \underbrace{H_t^{\parallel}}_{\text{medium 2}}$$

In terms of the magnetic fields ($\vec{B} = \mu \vec{H}$):

$$\frac{B_i^{\parallel}}{\mu_1} + \frac{B_r^{\parallel}}{\mu_1} = \frac{B_t^{\parallel}}{\mu_2}$$

Applying the same assumptions for the exponentials, we get the following result when plugging in the plane wave solutions for the magnetic fields:

$$\frac{B_{0i}}{\mu_1} \cos \theta_i + \frac{B_{0r}}{\mu_1} \cos \theta_i = \frac{B_{0t}}{\mu_2} \cos \theta_t$$

Now rewriting the magnetic field amplitudes in terms of electric field amplitudes:

$$\begin{aligned} \frac{|\vec{E}|}{|\vec{B}|} = v &\implies B_0 = \frac{E_0}{v} = E_0 \frac{n}{c} \\ \implies &\boxed{E_{0i} \frac{n_1}{\mu_1} \cos \theta_i + E_{0r} \frac{n_1}{\mu_1} \cos \theta_i = E_{0t} \frac{n_2}{\mu_2} \cos \theta_t \quad (\text{IC 4})} \end{aligned}$$

All that's left is a load of algebraic manipulation to solve for the reflection and transmission coefficients:

$$\begin{aligned}
E_{0t} &= E_{0i} - E_{0r} \\
E_{0t} &= \left(E_{0i} \frac{n_1}{\mu_1} \cos \theta_i + E_{0r} \frac{n_1}{\mu_1} \cos \theta_i \right) \frac{\mu_2}{n_2 \cos \theta_t} \\
\Rightarrow E_{0i} - E_{0r} &= E_{0i} \frac{n_1 \mu_2 \cos \theta_i}{n_2 \mu_1 \cos \theta_t} + E_{0r} \frac{n_1 \mu_2 \cos \theta_i}{n_2 \mu_1 \cos \theta_t} \\
E_{0i} \left(\frac{n_2 \mu_1 \cos \theta_t - n_1 \mu_2 \cos \theta_i}{\cancel{n_2 \mu_1 \cos \theta_t}} \right) &= E_{0r} \left(\frac{n_1 \mu_2 \cos \theta_i + n_2 \mu_1 \cos \theta_t}{\cancel{n_2 \mu_1 \cos \theta_t}} \right) \\
\frac{E_{0r}}{E_{0i}} &= \frac{n_2 \mu_1 \cos \theta_t - n_1 \mu_2 \cos \theta_i}{n_1 \mu_2 \cos \theta_i + n_2 \mu_1 \cos \theta_t} \left(\frac{1/\mu_1 \mu_2}{1/\mu_1 \mu_2} \right)
\end{aligned}$$

$$\boxed{\frac{E_{0r}}{E_{0i}} = -\frac{(n_1/\mu_1 \cos \theta_i - n_2/\mu_2 \cos \theta_t)}{(n_1/\mu_1 \cos \theta_i + n_2/\mu_2 \cos \theta_t)} \equiv r_{\perp} \quad (\text{reflection coefficient for S-polarization})}$$

Similarly, we can solve for the transmission coefficient:

$$\boxed{\frac{E_{0t}}{E_{0i}} = \frac{2(n_1/\mu_1) \cos \theta_i}{(n_1/\mu_1) \cos \theta_i + (n_2/\mu_2) \cos \theta_t} \equiv t_{\perp} \quad (\text{transmission coefficient for S-polarization})}$$

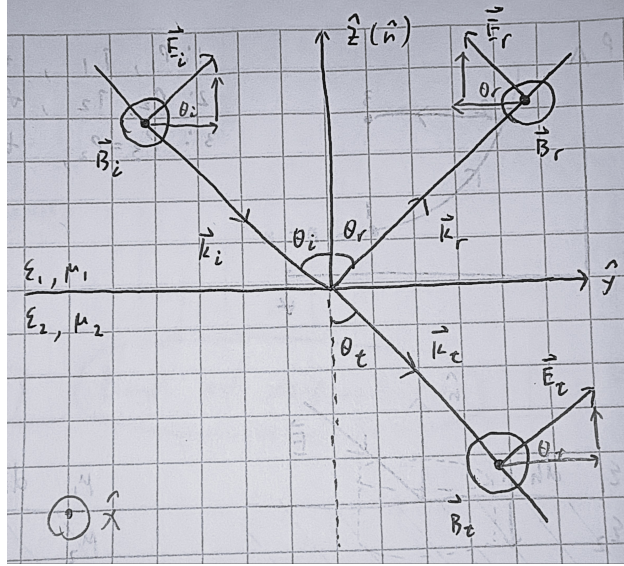
We can now predict the reflected and transmitted electric waves for any given incident electric wave (using the laws of optics and the Fresnel coefficients)!

Additionally, in most materials, the magnetic permeability doesn't change much, so we can further simplify by assuming $\mu_1 \approx \mu_2$:

$$\boxed{r_{\perp} = -\frac{\left(\cos \theta_i - \sqrt{(n_2/n_1)^2 - \sin^2 \theta_i} \right)}{\left(\cos \theta_i + \sqrt{(n_2/n_1)^2 - \sin^2 \theta_i} \right)}}$$

$$\boxed{t_{\perp} = \frac{2 \cos \theta_i}{\cos \theta_i + \sqrt{(n_2/n_1)^2 - \sin^2 \theta_i}}}$$

Case B: Fresnel Coefficients (P-Polarization)



Apply interface condition (3): $E_1^{\parallel} = E_2^{\parallel}$:

$$(E_{0i} e^{i(yk_i \sin \theta_i - \omega_i t)} \cos \theta_i) + (-E_{0r} e^{i(yk_r \sin \theta_r - \omega_r t)} \cos \theta_r) = (E_{0t} e^{i(yk_t \sin \theta_t - \omega_t t)} \cos \theta_t)$$

Simplify w/ laws of optics

$$\implies \boxed{E_{0i} \cos \theta_i - E_{0r} \cos \theta_i = E_{0t} \cos \theta_t \quad (\text{IC 3})}$$

Apply interface condition (4): $H_1^{\parallel} = H_2^{\parallel}$, $\vec{B} = \mu \vec{H}$

$$\frac{B_i^{\parallel}}{\mu_1} + \frac{B_r^{\parallel}}{\mu_1} = \frac{B_t^{\parallel}}{\mu_2} \xrightarrow{\text{simplify w/ laws of optics}} \frac{1}{\mu_1} B_{0i} + \frac{1}{\mu_1} B_{0r} = \frac{1}{\mu_2} B_{0t}$$

Rewrite to $\vec{E}$ using: $\frac{|\vec{E}|}{|\vec{B}|} = v_{\text{medium}} = \frac{c}{n} \implies B_0 = E_0 \frac{n}{c}$

$$\frac{1}{\mu_1} \left(\frac{n_1}{c} E_{0i} \right) + \frac{1}{\mu_1} \left(\frac{n_1}{c} E_{0r} \right) = \frac{1}{\mu_2} \left(\frac{n_2}{c} E_{0t} \right) \implies \boxed{\frac{n_1}{\mu_1} E_{0i} + \frac{n_1}{\mu_1} E_{0r} = \frac{n_2}{\mu_2} E_{0t} \quad (\text{IC 4})}$$

Find coefficients by solving for $\frac{E_{0r}}{E_{0i}} = r_{\parallel}$, $\frac{E_{0t}}{E_{0i}} = t_{\parallel}$:

$$E_{0i} \frac{\cos \theta_i}{\cos \theta_t} - E_{0r} \frac{\cos \theta_i}{\cos \theta_t} = E_{0t}$$

$$\implies \frac{n_1}{\mu_1} E_{0i} + \frac{n_1}{\mu_1} E_{0r} = \frac{n_2}{\mu_2} \left(E_{0i} \frac{\cos \theta_i}{\cos \theta_t} - E_{0r} \frac{\cos \theta_i}{\cos \theta_t} \right)$$

$$\implies E_{0r} \left(\frac{n_1}{\mu_1} + \frac{n_2 \cos \theta_i}{\mu_2 \cos \theta_t} \right) = E_{0i} \left(\frac{n_2 \cos \theta_i}{\mu_2 \cos \theta_t} - \frac{n_1}{\mu_1} \right)$$

$$\begin{aligned} \Rightarrow \frac{E_{0r}}{E_{0i}} &= \frac{\left(\frac{n_2 \cos \theta_i}{\mu_2 \cos \theta_t} - \frac{n_1}{\mu_1} \right)}{\left(\frac{n_1}{\mu_1} + \frac{n_2 \cos \theta_i}{\mu_2 \cos \theta_t} \right)} = \frac{\left(\frac{n_2 \cos \theta_i \cancel{\mu_1 \mu_2 \cos \theta_t}}{\cancel{\mu_2 \cos \theta_t}} - \frac{n_1 \cancel{\mu_1} \mu_2 \cos \theta_t}{\cancel{\mu_1}} \right)}{\left(\frac{n_1 \cancel{\mu_1} \mu_2 \cos \theta_t}{\cancel{\mu_1}} + \frac{n_2 \cos \theta_i \mu_1 \cancel{\mu_2 \cos \theta_t}}{\cancel{\mu_2 \cos \theta_t}} \right)} \\ \Rightarrow \boxed{\frac{E_{0r}}{E_{0i}} = \frac{n_2 \mu_1 \cos \theta_i - n_1 \mu_2 \cos \theta_t}{n_1 \mu_2 \cos \theta_t + n_2 \mu_1 \cos \theta_i} \equiv r_{\parallel}} &\quad \text{(reflection coefficient for P-polarization)} \end{aligned}$$

Finally, $\mu_1 \approx \mu_2$

$$\boxed{r_{\parallel} = \frac{E_{0r}}{E_{0i}} = \frac{n_2 \cos \theta_i - n_1 \cos \theta_t}{n_1 \cos \theta_t + n_2 \cos \theta_i}}$$

Now repeat for $t_{\parallel}$:

$$\begin{aligned} E_{0i} \cos \theta_i - E_{0t} \cos \theta_t &= E_{0r} \cos \theta_i \Rightarrow E_{0i} - E_{0t} \frac{\cos \theta_t}{\cos \theta_i} = E_{0r} \\ \Rightarrow \frac{n_1}{\mu_1} E_{0i} + \frac{n_1}{\mu_1} \left(E_{0i} - E_{0t} \frac{\cos \theta_t}{\cos \theta_i} \right) &= \frac{n_2}{\mu_2} E_{0t} \\ \Rightarrow E_{0i} \left(\frac{n_1}{\mu_1} + \frac{n_1}{\mu_1} \right) &= E_{0t} \left(\frac{n_2}{\mu_2} + \frac{n_1 \cos \theta_t}{\mu_1 \cos \theta_i} \right) \\ \Rightarrow E_{0i} \left(\frac{2n_1 \cancel{\mu_1} \mu_2 \cos \theta_i}{\cancel{\mu_1}} \right) &= E_{0t} \left(\frac{n_2 \mu_1 \cancel{\mu_2 \cos \theta_i}}{\cancel{\mu_2}} + \frac{n_1 \cos \theta_t \cancel{\mu_1} \mu_2 \cos \theta_i}{\cancel{\mu_1 \cos \theta_i}} \right) \\ \Rightarrow E_{0i} (2n_1 \mu_2 \cos \theta_i) &= E_{0t} (n_2 \mu_1 \cos \theta_i + n_1 \mu_2 \cos \theta_t) \\ \Rightarrow \boxed{\frac{E_{0t}}{E_{0i}} = \frac{2n_1 \mu_2 \cos \theta_i}{n_2 \mu_1 \cos \theta_i + n_1 \mu_2 \cos \theta_t} \equiv t_{\parallel}} &\quad \text{(transmission coefficient for P-polarization)} \end{aligned}$$

Set $\mu_1 \approx \mu_2$

$$\boxed{t_{\parallel} = \frac{E_{0t}}{E_{0i}} = \frac{2n_1 \cos \theta_i}{n_2 \cos \theta_i + n_1 \cos \theta_t}}$$

In summary, we can use the laws of optics to determine the directions of the reflected and transmitted waves, and then we can use the Fresnel coefficients to determine the amplitudes of the reflected and transmitted waves. Altogether, we can derive the plane wave solutions for the reflected and transmitted waves based on the plane wave solution for the incident wave and thus can do all the normal wave calculations for the reflected and transmitted waves (ex: energy flux, intensity, etc.).

Reflectance and Transmittance

Total Internal Reflection

Evanescent Waves

Brewster's Angle